The Hidden Costs of Urban Renewal[*]

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Abstract

Three constraints hinder urban redevelopment: (i) market structure (concentrated developers); (ii) market frictions (uncertainty regarding future sales prices, clearing costs, and construction time); and (iii) market regulation (taxes and density caps). I formalise these constraints within an option value model of parcel redevelopment, in which land is traded between landowners and oligopsonistic developers. I calibrate the model on data from 550,000 parcels in the Paris metropolitan area, recover a parcel-specific fixed clearing cost net of the other wedges, and compute medium-run (20-year) supply responses under alternative scenarios. The model identifies 53,642 parcels with redevelopment potential. Lifting the density cap raises the quality-adjusted housing stock by 0.27% after twenty years. A self-financing fixed-cost subsidy raises it by 3.80% and reduces regional rent by 5.41%. Restoring perfect competition on the land market raises it by 1.09% and brings redevelopment forward by 3.8 years relative to the *laissez-faire* equilibrium. Exploring alternative policy options, I find that a tax on under-used land can achieve similar magnitudes. Finally, I examine which subsidies could help local authorities to increase their share of controlled-rent housing.

Keywords: Redevelopment, monopsony, concentration, fixed costs, real options, housing supply. **JEL**: R14, R31, R33, D43, L13, H23

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1. Introduction

In the large cities of rich countries, development on the outskirts is limited by technical constraints on commuting time, the urban fringes of nearby cities, or regulations on land preservation.[1] Between 1999 and 2021 the population of the Paris area grew by 11%, while its built-up land grew by less than 1% (Figure H1). About three quarters of the housing units built since 2014 replaced a structure demolished on the same parcel. The pattern is similar in New York, where two thirds of residential permits in 2023–2025 were filed on parcels that already carried a building (Lebret et al., 2026). Even when housing costs are high, and keep increasing, the supply response is low (Saiz, 2010; Baum-Snow and Han, 2024) and slow (Rollet, 2025). In the Paris metropolitan area, around 4,700 parcels zoned for housing are empty, vacant, or far below the height of their neighbours, even as they sit where rents are among the highest in France. Redevelopment is not only rare, it is also slow, and this paper asks why, and what policy can do about it.

Slow redevelopment is not necessarily caused by an inefficient allocation. In fact, even a social planner, facing uncertain future prices, long time-to-build, and fixed clearing costs, might also delay or restrict construction. If the market is competitive, developers and social planners redevelop at the same time (Leahy, 1993). In this paper, I study the urban redevelopment market by explicitly modelling the interplay between three essential wedges. Market structure, when developers are concentrated, lets them pay owners less than their land is worth. Owners respond by selling later, and some never sell. Market frictions (uncertainty about future prices, clearing costs, and construction time) create an option value of waiting that delays redevelopment, for private parties and for a planner alike. Market regulations, such as taxes and density caps, reduce the profitability of redevelopment. Taxes vary across municipalities and caps across cadastral sections. I show that these wedges can have very different effects on housing supply, rents, development timing, and allocative efficiency.

I leverage a granular parcel-level dataset of the Paris metropolitan area, linking tax records, building envelopes, and the universe of building permits and land sales. For each parcel, I observe the standing structure and its floorspace composition between housing and other uses, the rents and taxes it generates, ownership, applicable land-use rules, and soil condition. Prompted by novel empirical evidence on market concentration (the 7 largest developers supply 16% of redeveloped floorspace), construction project lengths (3–4 years on average), and land-use reallocation, I set up and calibrate a dynamic model

[1] The European Union targets net-zero land take by 2050 (European Commission, 2021). Numerous cities have long imposed urban growth boundaries, see for instance Koster (2023) for an analysis of "greenbelts" in the United Kingdom.

of parcel redevelopment featuring land market oligopsony, construction frictions, and land-use regulations. I theoretically and quantitatively show that these wedges hinder construction and that targeted policies can restore allocative efficiency. Oligopsony on the land market lowers the price that landowners receive, so they sell less often and keep their parcels longer. Construction frictions and permitting delays raise the cost and the time needed to build, which reduces the expected return on redevelopment. Developers therefore postpone or abandon projects that would be efficient to undertake.

Buildings already exist on parcels and generate income for the incumbent landowners. If the net present value of a redevelopment project exceeds the current revenue flow of the parcel, a developer can buy the parcel, tear down the existing structure, and build a new, denser building on it. A small number of developers make offers to owners, who differ in their land price valuation, which remains unobserved by the developers. Because a few agents have a large market share, the market faces the typical monopsony problem: a developer will offer less than the parcel is worth, since the supply of parcels is upward-sloping. I estimate this monopsony markdown, which rises with local concentration, using methods borrowed from the labour literature.

Markdowns distort both redevelopment timing and its spatial distribution. A landowner who sells a parcel receives only a fraction of what the land is worth to the developer. She behaves as if her building were worth more than it truly is, and waits longer before accepting lower offers. Concentration therefore delays redevelopment, and can block it entirely on valuable standing structures. Since concentration and building values vary across parcels, the markdown reorders which sites are redeveloped, potentially leading to spatial misallocation of projects.

Three further wedges hinder redevelopment: construction takes time (two to three years on average), demolition is irreversible and costly, and density caps and taxes constrain profitability. These cost and regulatory wedges create an option to wait for higher rents, which delays redevelopment even under perfect competition. Unlike the markdown, these frictions would also constrain a social planner. They raise the planner's thresholds as well as the private ones, so they delay redevelopment without changing the order in which the planner would convert parcels. The markdown enters only the private threshold, and it varies with the value of standing structures. It therefore changes which parcels are redeveloped, how many, and in which order. If the planner counts tax revenue as social value, taxes and caps misallocate redevelopment too, because they vary across municipalities and sections. I measure both forms of misallocation.

The combination of administrative and construction data allows me to identify concentration, measure

the standing structure value, and estimate novel monopsony markdowns. I estimate it from sales of land that are followed by a permit on the same parcel, which reveal both the price and the identity of the buyer. The price a developer pays, relative to what the parcel is worth to her, falls with her local market share. The slope of this relation measures how fast the markdown rises with market share. The operating margins of listed developers pin down its level. Together they identify the two elasticities. Additionally, I can recover the fixed cost of redeveloping a parcel from the price of land. A parcel's price equals the value of what stands on it, plus the owner's share of the option to redevelop it. Once the markdown is known, the price reveals the option, and the option reveals the cost. The model is never fitted to when parcels are redeveloped, so the observed timing is a test. Among parcels the model classifies as ripe, 0.9% receive a permit, against 1.8% on average. The model is compared empirical moments such as the elasticity of housing supply and how land prices vary with concentration across municipalities.

The model calibration shows how strongly each wedge constrains supply. Landowners receive on average 62% of the value their land has for developers. Under atomistic competition, with many small developers, this share would rise to 100%. I define the supply shortfall as the reduction in cumulative floorspace redeveloped over twenty years, relative to a benchmark with no wedge. This benchmark is extreme, since new floorspace is 17.2 times the private level. In a Shapley decomposition, 53.5% of the shortfall comes from the fixed cost alone, 30.0% from taxes and density caps, 68.5% from frictions in total, and 1.5% from the markdown. Removing the fixed cost alone closes 48.9% of the gap. Removing taxes or caps alone closes 9.1% and 0.4%, because their effect depends on the other wedges. The median parcel with a standing structure becomes ripe 1.8 years later than a planner would allow. The shares leave out the 112,299 parcels that concentration blocks for ever, so the 1.5% is a lower bound on its cost. Concentration reorders redevelopment across parcels, but the effect on quality-adjusted supply is small and negative. Over twenty years, 9% of new floorspace is built on parcels that a social planner would have converted only later. If total floorspace is held fixed but built in the planner's order, the change in quality-adjusted floorspace is -8% of the gap between the private economy and the planner.

I use the calibrated model to evaluate five policies: the first removes the density cap, the second is a self-financing subsidy to the fixed cost, the third restores a competitive land market, which is what a planner would ensure, and the fourth implements the third with a subsidy to sellers. All results are in general equilibrium: more supply lowers rents, which feeds back into the decision to build. Every policy faces the same path of demand, so differences come from the policies and not from luck. For each policy I

report four results: the change in the quality-adjusted housing stock and in rents after twenty years, the net cost to the public budget per additional housing unit, the time the policy saves (the number of years by which it brings forward the stock the baseline reaches in twenty years), and where the new housing is built.

Removing the density cap raises the quality-adjusted housing stock by 0.27% and lowers rents by 0.40% over twenty years. It delivers the stock of year twenty 1.0 years earlier. Subsidising fixed clearing costs at the largest self-financing scale, 1,050€ per square metre of land, raises the stock by 3.80% and saves 16.2 years. Restoring competition in the land market raises the stock by 1.09%, lowers rents by 1.60%, and saves 3.8 years. Under the current calibration, breaking up developers can fully restore a competitive outcome. A subsidy to sellers can also fully close this gap. Developers pay landowners less than the competitive price, and this gap is the same share of the price across parcels within a municipality. The local Herfindahl index measures how concentrated developers are, and therefore how large this gap is. A subsidy set at that index raises the price owners receive back to the competitive level on every parcel. Owners then sell when they would under perfect competition. The planner needs no data on each parcel's clearing costs or on individual owners. The net cost is 99.9 thousand euros per additional housing unit, in the same range as the direct cost of building.

I study two further policies. A tax on under-used land, which achieves similar results as the fixed cost subsidies. I also examine which subsidies would help local authorities raise their share of controlled-rent housing, through social housing mandate (a carrot-and-stick policy, as in (Lebret et al., 2026)).

Targeted policies can ease housing crises in large cities even when real frictions remain. The key is to separate market failures, such as concentration, from frictions that any builder would face. A calibrated subsidy against local market concentration needs no parcel-level information. It restores allocative efficiency on the land market.

Related literature The paper contributes to four literatures. First, it changes how the cost of supplying housing in built-up cities is measured. A long tradition measures regulation as the gap between prices and construction costs (Glaeser and Gyourko, 2003; Gyourko and Molloy, 2015), and relates supply elasticities to geography and land availability (Saiz, 2010; Baum-Snow and Han, 2024). A macroeconomic literature shows that housing constraints misallocate workers across cities (Hsieh and Moretti, 2019; Herkenhoff et al., 2017; Ganong and Shoag, 2017; Glaeser and Gyourko, 2018), extending the analysis of misallocation across firms (Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008). Land is also misal-

located across uses in developing countries (Duranton et al., 2015). I study misallocation within a city, across parcels. I show which wedges change the order in which parcels are converted. Other wedges and costs only lower or delay supply. The theory of redevelopment goes back to Brueckner (1980) and Rosenthal and Helsley (1994). Work on teardowns documents when the option to replace a structure is exercised (Dye and McMillen, 2007; McMillen and O'Sullivan, 2013). Recent structural models recover large fixed costs by matching the observed pace of construction (Murphy, 2018; Peng, 2023; Rollet, 2025). Combes et al. (2021) and Chapelle et al. (2023) measure housing production and land-use regulation in France. I recover the fixed cost from a different moment, the price of land. The redevelopment rate then tests the model instead of identifying it. The cost also comes net of the delay, the developer's markdown, and taxes. A quantity-based approach folds these into a single number.

Second, the paper brings buyer power into the real-options theory of land development (Titman, 1985; McDonald and Siegel, 1986; Capozza and Helsley, 1990; Williams, 1991; Dixit and Pindyck, 1994), its empirical tests (Quigg, 1993; Cunningham, 2006), and its extensions to time to build (Majd and Pindyck, 1987; Bar-Ilan and Strange, 1996; Oh et al., 2024) and to predictable demand growth (Lange and Teulings, 2024). In this literature, market structure works through strategic exercise. Rivals holding options on similar projects build earlier and in clusters (Grenadier, 1996, 2002). Competition erodes the value of waiting (Bulan et al., 2009). My mechanism works on the other side of the transaction. Developers compete to buy the option from owners. Their buyer power raises the threshold at which owners of a standing structure are willing to sell. The closest paper is Lebret et al. (2026), who study redevelopment incentives with a real-options model of New York. I add how land changes hands, recover costs parcel by parcel from prices, and separate the market failure from the frictions. Moszkowski and Stackman (2023) study the role of option value and owner heterogeneity in explaining vacancies of commercial real estate.

Third, the paper takes the modern analysis of monopsony to the land market. Monopsony goes back to Robinson (1933). Search is one of its sources (Burdett and Mortensen, 1998; Manning, 2003; Bhaskar et al., 2002). Recent work makes it estimable. Firms face upward-sloping labour supply because workers have idiosyncratic preferences over employers (Card et al., 2018; Lamadon et al., 2022). With few employers per market, the markdown depends on concentration (Berger et al., 2022; Azar et al., 2022; Yeh et al., 2022), mirroring the product-market model of Atkeson and Burstein (2008). Kroft et al. (2025) show that construction firms hold power in both labour and product markets. In real estate, market power has been studied mostly on the seller side. Concentrated landlords raise rents in New York (Watson and

Ziv, 2026). Developers charge large markups in São Paulo (Alvarez and Biderman, 2026). Homebuilding is concentrated in the United States (Quintero, 2023; Favilukis et al., 2026) and in France (Lefebvre and Daly, 2026; Charreire et al., 2025). Markusen and Scheffman (1977) ask whether developers held buyer power in the Toronto land market. Land assembly creates hold-out rents (Miceli and Sirmans, 2007; Brooks and Lutz, 2016). Search models of the housing market offer another channel (Wheaton, 1990; Genesove and Han, 2012; Piazzesi et al., 2020). Under efficient bargaining, however, concentration lengthens the search for a buyer. It leaves the decision to redevelop unchanged. I show that buyer power combined with private owner values distorts the decision itself. I estimate it from land transactions.

Finally, the paper ranks supply-side housing policies by whether they need a fiscal justification at all. Housing subsidies are largely capitalised into prices (Fack, 2006). Subsidised construction partly crowds out unsubsidised construction (Eriksen and Rosenthal, 2010). Inclusionary mandates act as a tax on development (Schuetz et al., 2011; Soltas, 2024). I evaluate subsidies, taxes, density relaxations, and inclusionary mandates in one framework. Costs and receipts are counted over the same horizon. I also derive the instrument that corrects the market failure. Under labour-market monopsony, a minimum wage can restore employment (Robinson, 1933; Manning, 2003). Its land-market analogue, a floor on land prices, would eliminate developers' rents and their entry. A subsidy to sellers restores efficient timing. It leaves developers' rents in place.

Roadmap Section 2 presents the setting, the data, and the facts that motivate the model. Section 3 develops the model, its welfare properties, and its identification. Section 4 calibrates it and estimates the markdown. Section 5 validates the model, decomposes the supply shortfall, and measures misallocation. Section 6 evaluates the policies. Section 7 concludes.

2. Setting, data and facts

2.1. Institutional context

The analysis is set in the Paris area: the city of Paris (*département* 75) and the three contiguous *départements* of Hauts-de-Seine (92), Seine-Saint-Denis (93), and Val-de-Marne (94), together known as the *Petite couronne*. The area is dense. All but two of its 143 municipalities, including the 20 Paris *arrondissements*, fall in the highest INSEE density class. The area hosts around 10% of the French population and produces 20% of French GDP. Demand for floorspace is strong and puts sustained pressure on both the commercial and the residential markets.

Local governments levy taxes and control land use. Mayors deliver building permits (Chapelle et al., 2023, 2025). The zoning plan is the *plan local d'urbanisme* (PLU). In Paris, the city adopts it. In the three other *départements*, the territorial establishments (EPT) of the *Métropole du Grand Paris* have held the competence since 2016 and draft intercommunal plans (PLUi). Permitting is partly discretionary: a permit that complies with the plan can still be refused. The applicant bears the delay of instruction and the risk of refusal. As a rule, land and the building on it are sold together. A developer who wants a parcel must buy the standing structure on it, and its owner is compensated for the value she gives up. When a building is held in co-ownership, selling it as a whole requires the agreement of every owner. When it is rented, the landlord can end a lease to sell or rebuild only at its term and with notice (*loi du 6 juillet 1989*).

A developer rarely buys land outright. She first signs a conditional sale agreement (*promesse de vente*) at a fixed price. The agreement lapses if no permit is obtained or if the permit is successfully challenged. The deed is signed once the permit is final. The developer therefore holds an option on the parcel before she owns it, and the deed recorded in DVF usually follows the permit. The model abstracts from this two-step transaction. It treats the date at which the owner commits as the start of the project (Section 4.1).

A developer is a firm that prospects and buys land and supplies new housing units on it. Developers are identified by their SIREN number, the identifier the French business register assigns to every legal entity, including the project-specific vehicles that many developers incorporate for a single redevelopment. I focus on private developers. Public developers supply a significant share of housing, but they do not face the same constraints and do not take part in the land market in the same way.

Since 2000, municipalities that fall short of a target share of social housing in their stock (20 to 25%, depending on local circumstances) must write inclusionary requirements into their PLU under the *loi SRU*, and face a financial penalty for missing the target.

2.2. Data

I assemble a parcel-level panel for the Paris area from 2014 to 2025. The *Fichiers Fonciers* provide the universe of taxed units, with construction date, floorspace, and destination (housing, tertiary, secondary, or annex). I aggregate units to the parcel and assign a destination once at least 90% of a parcel's units share it, calling the parcel mixed use otherwise. *Sitadel*, the universe of authorised building permits, identifies new developments, their floorspace and unit count, the issuance date, the applicant, and, for about 60% of permits, the construction start and end dates. I retain permits that add at least one housing

unit of at least 10 square metres and whose applicant is a private firm or a natural person, which excludes pure social-housing operations. Sale prices come from the universe of land and property transactions (*Demande de Valeurs Foncières*, DVF). Rents come from the *Observatoire des Loyers*, a hedonic municipal rent index available from 2015. The *BD Topo* gives the three-dimensional envelope of every building in the region in successive vintages, which distinguishes demolitions and additions from measurement noise. Land-use zoning comes from *Apur*, population and income from *Fidéli*, and ground condition and utility networks from the BRGM and network operators. Demolition and construction costs come from the *Prix de revient des logements neufs* survey (PRLN), a stratified survey of realised project costs.

When the applicant is a firm, the permit is matched to a developer through its SIREN (Appendix Table A3 reports match rates). I consolidate project-specific vehicles into their parent group where the parent can be identified (Appendix A.7). A subset of DVF land sales is followed by a permit on the same cadastral parcel. For these sales I observe both the price and the identity of the buyer, which is what identifies the markdown (Section 4.2).

2.3. Facts

Fact 1. Redevelopment is the supply margin Matching the universe of permits creating housing (2014–2024) with the building envelope before and after issuance identifies three types of construction: new units on empty parcels, transformation within an existing envelope, and demolition-reconstruction with a change in the envelope. Over the period, 18,873 permits created 298,329 units and 18.0 million square metres of residential floorspace. About three quarters of the new units come from demolition-reconstruction, about a fifth from within-envelope transformation, and the rest, under 5%, from empty parcels. Figure 1 breaks these shares down by the use of origin of the parcel. In the universe of permits, including those that create no housing, 69% create housing. The share is 79% for permits that involve a demolition and 56% for the others.

Fact 2. Redevelopment reallocates land across uses, and demolishing non-housing structures yields the densest projects Mixed-use parcels deliver about two fifths of new units, and former tertiary-activity parcels about a sixth (Figure 1). Demolitions of tertiary, secondary, mixed-use, and multi-family structures have median floor-area ratios between 0.78 and 1.76. Every other project type, including the demolition of single-family houses and annexes, has a median ratio between 0.21 and 0.62. Demolishing a tertiary building is the densest form of redevelopment per square metre of land, with a median ratio

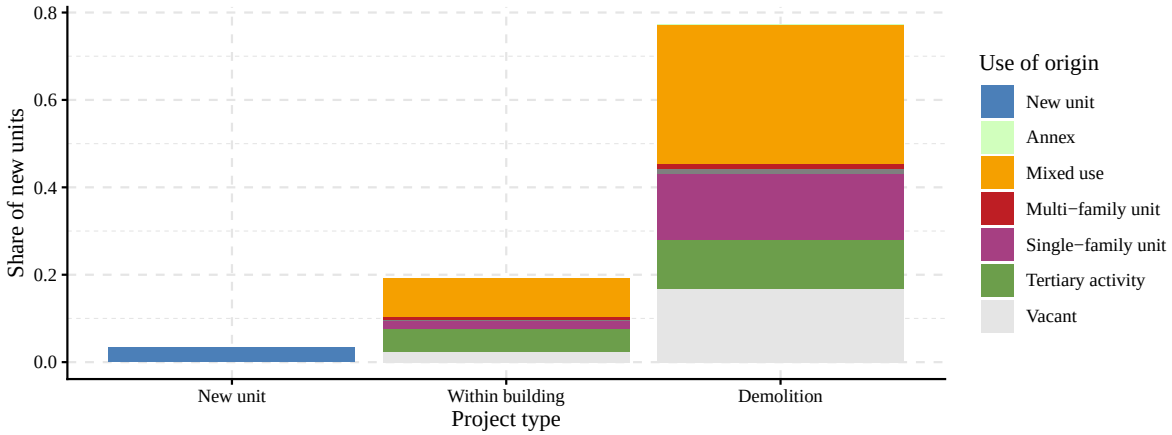


Figure 1: Housing supply by type of transformation and use of origin

Data: *Sitadel*, *Fichiers Fonciers*, *BD Topo* (2014–2025). New units are housing units built on empty lots. Within building are housing units provided in buildings without change in envelope. Demolition are housing units provided in demolished and rebuilt buildings. Use of origin is the main occupation of the parcel before the permit. Shares of new units by transition type $\times$ use of origin.

of 1.76 (Table G1).

Fact 3. Construction time depends on the use of origin Once work starts, tertiary and secondary origins are the slowest to build on, with medians of 2.2 to 2.7 years. Housing, annex, mixed, and vacant origins take 1.4 to 2.3 years. For five of the seven origins, construction takes longer after a demolition than within an existing envelope. From acquisition to completion, projects take 3.5 to 6.5 years. Demolitions of secondary-activity and mixed-use parcels take longest, and new construction on empty parcels is among the fastest, at 3.7 years. Figure 2 plots the medians. Appendix Table G2 tests the differences in construction time: tertiary origins differ from every other origin except secondary activity. This motivates a construction lag indexed by the use of origin (Assumption 3).

Fact 4. Individual projects do not move rents around them No single project is large relative to the housing stock. The largest project in a given municipality represents a median of 1.35% of its dwelling stock. To assess whether projects change prices around them, I run a donut difference-in-differences around new projects, by project size. Because no micro-data exist for rents, I use the price per square metre of existing dwellings resold nearby, recorded in DVF. For each project, I compare the change in prices before and after delivery in an inner ring around its footprint with the change in an outer ring around the same project. A narrow band close enough to pick up construction nuisance or the new units themselves is excluded. The design follows the literature on the local effects of new buildings (Asquith

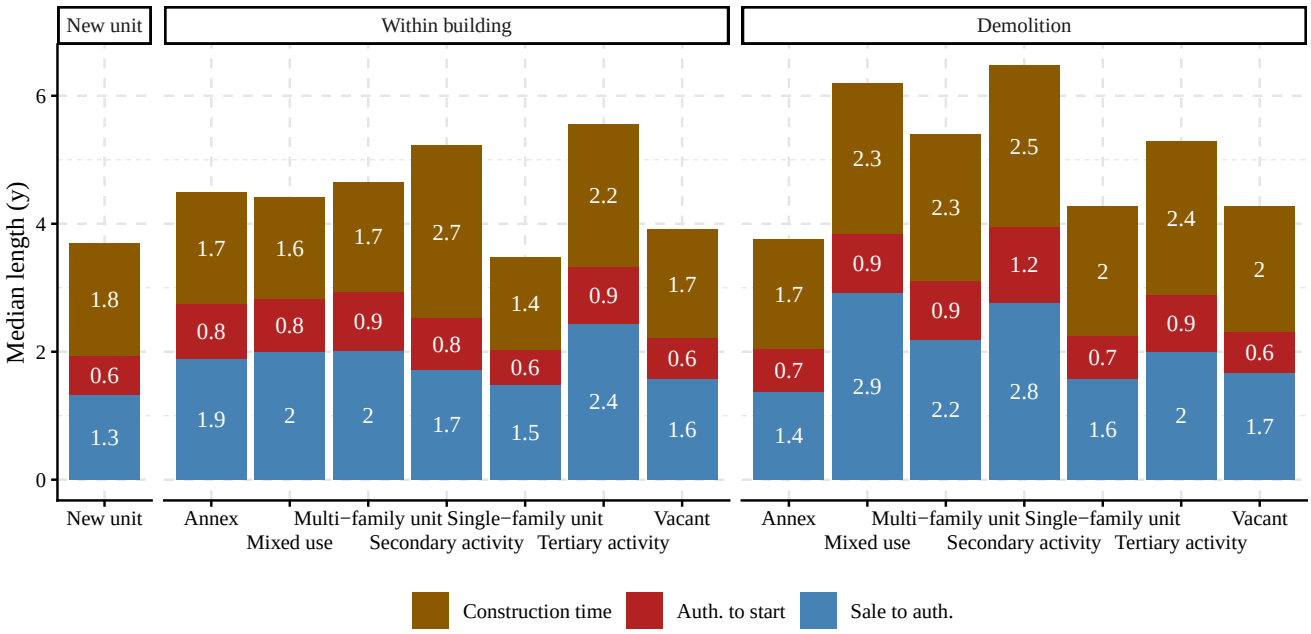


Figure 2: Construction time across uses of origin

Data: *Sitadel*. Median time from parcel acquisition to permit authorisation ("Sale to auth."), from authorisation to the start of works ("Auth. to start"), and from the start of works to completion ("Construction time"), by project type (new unit, within-building transformation, and demolition) and use of origin, for permits creating at least one housing unit.

et al., 2023; Li, 2022; Pennington, 2021), which finds that large new buildings lower nearby rents by a few percent in dense US cities. Appendix A.8 gives the specification. For projects of at least 50 dwellings, the estimated effect on nearby prices is 0.076 (s.e. 0.057). The estimates for all projects and for the other size classes have either sign, and none is significant at conventional levels (Appendix Table A1). They are imprecise, and their confidence intervals do not rule out effects of the size found in US cities. The case for price-taking therefore rests mainly on the size of projects relative to the stock. Figure H3 plots the event study by project size and by year relative to delivery.

Fact 5. Land is bought by a handful of developers Acquiring the parcel that supplies a project is a different matter from selling the housing it delivers. Developers do not take the price of a parcel as given the way they take rents as given (Fact 4). Out of 3,977 developers active over 2013–2024, the top seven account for 16% of redeveloped floorspace. The region-wide Herfindahl index of floorspace shares is 0.007, which corresponds to 153 equally sized developers. Concentration is higher where land actually changes hands. Within a municipality, the median effective number of developer groups is 3.0, and the largest group holds a median share of 50%. Figure 3 plots the concentration curve, and Table G3 reports shares and counts. The correlation between the municipal Herfindahl index and municipal rents

is -0.06. This correlation matters for whether concentration distorts supply in the most or in the least valuable locations (Section 5.3). These figures understate concentration, because the consolidation of project vehicles into their parent groups is still incomplete (Appendix A.7).

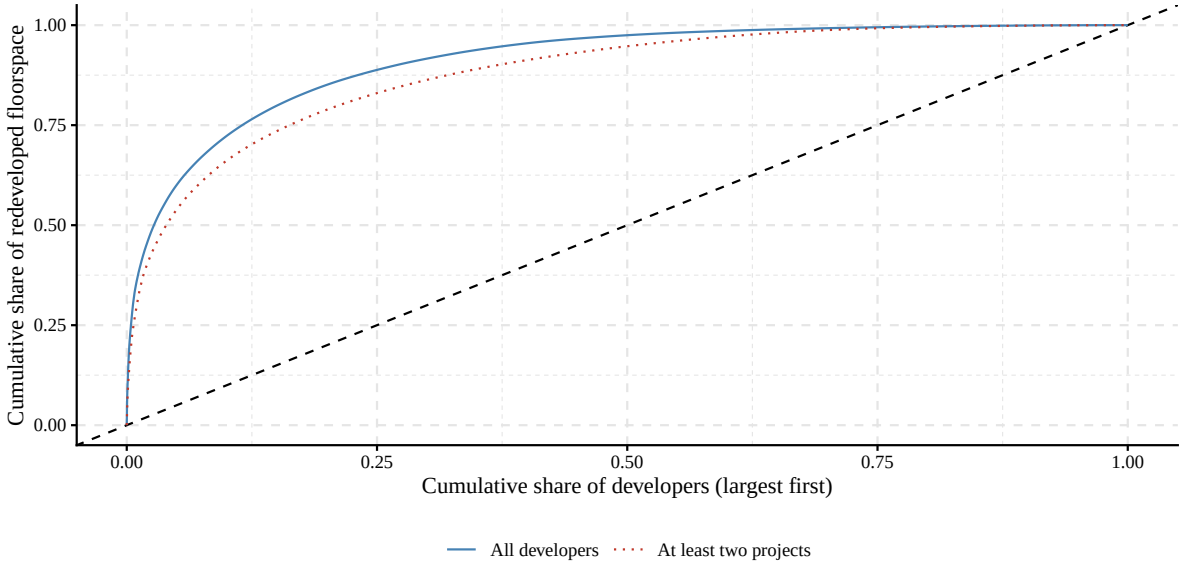


Figure 3: Region-wide developer concentration

Data: *Sitadel*, developer SIREN identifiers, 2013–2024. Cumulative share of redeveloped floorspace against the cumulative share of developers, ranked largest first. The dashed 45° line denotes perfect equality. The plain blue line is all projects with a recorded developer. The red dotted line keeps only developers who handled multiple projects during the period.

Fact 6. A large number of parcels remain idle Three types of parcel in areas zoned for housing could be redeveloped. Empty parcels host no physical building. Vacant parcels host a building but no taxed unit, and I retain only those with a documented past economic activity, to limit false positives. Infill parcels host a building significantly lower than its neighbours within 100 metres. Table 1 shows that redeveloping them to the density of their surroundings would add about 20 million square metres and 333,000 units across 4,700 parcels. Empty and vacant parcels concentrate in the closest suburbs of Paris (Figure H2), in the 98th percentile of French rents, and most have not changed in a decade. That they are not redeveloped at those rents is the puzzle the rest of the paper explains.

3. A model of redevelopment

The model is a parcel-level real-options problem in the tradition of McDonald and Siegel (1986), Capozza and Helsley (1990), and Dixit and Pindyck (1994). Timing is driven by rents rather than by idiosyncratic shocks to profits or costs. Construction takes time. Land changes hands in an oligopsonistic market,

Table 1: Potential housing supply on empty, vacant and infill parcels

Type	Parcels	Potential floorspace	Potential units
Empty	971	730,231	12,171
Infill	2,110	14,247,115	237,452
Vacant	1,608	5,035,369	83,923
Total	4,689	20,012,715	333,546

Data: *Fichiers Fonciers*, *BD Topo* (2024). Parcels are located in residentially zoned areas. Potential floorspace is the median building-coverage ratio × parcel area × median height within 100 metres, divided by a 2.5-metre ceiling height. Potential units assume 60 square metres per unit.

where a few developers make offers to owners whose value of keeping their parcel is private. I describe the environment, the project, the land market, the timing decision, the planner, what the price of a parcel reveals about its fixed cost, and how parcels add up to a supply path.

3.1. Environment and demand

A continuum of parcels is indexed by i. Each has an area T_i, a use of origin $u(i)$ (housing, mixed, non-housing, or vacant), a standing structure, a regulatory density limit $\bar{h}_i$, and a municipality $m(i)$, written m when there is no ambiguity. Municipality m has a location quality ϕ_m. Time is continuous and every agent discounts at rate ρ.

Assumption 1 (Demand)**.** A demand shifter A_t follows a geometric Brownian motion, $dA_t = \mu A_t\,dt + \sigma A_t\,dW_t$, with drift μ and volatility $\sigma > 0$. Rents clear the market against the delivered stock of quality-adjusted housing floorspace, $Q_t = \sum_i \phi_m h_{it}$, where h_{it} is the housing floorspace on parcel i at date t: $R_t = A_t Q_t^{-\alpha}$, where $\alpha > 0$ is the inverse elasticity of demand. The rent per square metre on parcel i is $R_{it} = \phi_m R_t$.

Weighting floorspace by location quality gives the model a spatial dimension: a square metre built where ϕ_m is high accommodates more demand than a square metre built where it is low, because households value it more. Every parcel faces the local rent $\phi_m R_t$.

Assumption 2 (Perceived rent process)**.** Each parcel has measure zero. Landowners and developers treat the rent as a geometric Brownian motion with drift $\delta < \rho$ and volatility σ: $dR_t = \delta R_t\,dt + \sigma R_t\,dW_t$.

No agent internalises her own effect on supply, which Fact 4 supports for an individual project. The drift δ summarises the growth of rents net of the erosion the market expects from future supply. I calibrate it from the net residential yield. From here on, R denotes the local rent of the parcel under consideration.

Standing structure Until it is redeveloped, parcel i yields the rent of its standing housing floorspace, $R\,h_i^{\text{old}}$, and a non-housing rental flow v_i^C. The property tax on the standing structure is not deducted from these flows (Appendix C.5). Housing rents follow the local rent. Commercial leases (*baux commerciaux*) revise rents only with a lease index, so I treat the non-housing flow as fixed. The value of keeping the structure in place, which I call the standing value, is

$$V_i^w(R) = \frac{R\,h_i^{\text{old}}}{\rho - \delta} + \frac{v_i^C}{\rho}, \tag{1}$$

with $h_i^{\text{old}} = 0$ for non-housing parcels, $v_i^C = 0$ for housing parcels, and $V_i^w = 0$ for vacant parcels.

3.2. Taxes

Four taxes enter the model. Value-added tax, at rate τ^V (20% for new housing), is levied on the sale price of new units. Input VAT is deductible, so VAT is a proportional wedge on the value of the completed building. The property tax (*taxe foncière*) is paid annually by the owner of the completed building after an exemption of e years, at a municipal rate t_m^F on a base equal to a fraction ψ^{new} of market rent. It is capitalised into the value of new units at the effective rate $\tilde{t}_m^F \equiv t_m^F \psi^{\text{new}} e^{-(\rho-\delta)e}$. I drop the subscript m when there is no ambiguity. The development tax (*taxe d'aménagement*), τ^A per square metre of new floorspace, is paid at the permit date. The transfer tax (*droits de mutation à titre onéreux*, DMTO), at rate τ^M, is paid by the buyer on the price of the land. VAT and the property tax enter the value of the project. The development tax and the DMTO enter the developer's value of the land.

3.3. The project

Assumption 3 (Technology and time to build)**.** A project started on parcel i at date t delivers h square metres of floorspace at $t + \ell_i$, where the lag $\ell_i = \ell_{u(i)}$ depends on the use of origin (Fact 3). Density is chosen at the start date and is bounded by the cap $\bar{h}_i$. The project cannot be abandoned once started. The standing structure is emptied at the start date. The fixed cost F_i is sunk at the start. Variable construction cost, net of deductible VAT, is $c_0 T_i^{-1/\zeta} h^{1+1/\zeta}$, a Cobb-Douglas technology in land and capital following Combes et al. (2021), spread evenly over the build.

Two discount factors summarise the lag. Rental income starts at completion, so its present value is scaled by $\lambda_i^R = e^{-(\rho-\delta)\ell_i}$. Construction outlays are spread over the build, so their present value is scaled by $\lambda_i^C = (1 - e^{-\rho\ell_i})/(\rho\ell_i)$. Their ratio $\kappa_i \equiv \lambda_i^R / \lambda_i^C$ is the effective discount on the net cash flow.

Lemma 1 (The lag rescales the payoff). *Let $\pi_i^{0,\tau}$ denote the payoff without lag. For every parcel and every rent, $\pi_i(R) = \lambda_i^C \, \pi_i^{0,\tau}(\kappa_i R)$. Proof in Appendix B.*

The developer's payoff at rent R, gross of the fixed cost and net of taxes on the completed building, is

$$\pi_i(R) = \max_{h \le \bar{h}_i} \frac{1-\tilde{t}^F}{1+\tau^V} \lambda_i^R \frac{R\,h}{\rho-\delta} - \lambda_i^C \, c_0 T_i^{-1/\zeta} h^{1+1/\zeta}. \tag{2}$$

Write $\hat{\kappa}_i \equiv \kappa_i(1-\tilde{t}^F)/(1+\tau^V)$ for the effective discount once the two taxes on the building are included. Ignoring the cap, the first-order condition gives the density the developer would choose at rent R. It increases with R and exhausts the cap at the saturation rent $\bar{R}_i$:

$$h_i^\star(R) = \left[\frac{\zeta\,\hat{\kappa}_i R}{c_0(\zeta+1)(\rho-\delta)}\right]^\zeta T_i, \qquad \bar{R}_i = \frac{1}{\hat{\kappa}_i} \cdot \frac{c_0(\zeta+1)(\rho-\delta)}{\zeta} \left(\frac{\bar{h}_i}{T_i}\right)^{1/\zeta}. \tag{3}$$

Below $\bar{R}_i$, in the interior regime, the developer builds $h_i^\star(R)$. Taxes lower density like a higher construction cost, since $h_i^\star$ is proportional to $(\hat{\kappa}_i/c_0)^\zeta$. Above $\bar{R}_i$, in the constrained regime, she builds the cap $\bar{h}_i$. The payoff is

$$\pi_i(R) = \begin{cases} \dfrac{\lambda_i^C}{\zeta}\left[\dfrac{\zeta\,\hat{\kappa}_i R}{(\zeta+1)(\rho-\delta)}\right]^{\zeta+1} c_0^{-\zeta} T_i & \text{if } R < \bar{R}_i \quad \text{(interior)} \\ \dfrac{1-\tilde{t}^F}{1+\tau^V}\lambda_i^R \dfrac{R\,\bar{h}_i}{\rho-\delta} - C_i & \text{if } R \ge \bar{R}_i \quad \text{(constrained)}, \end{cases} \tag{4}$$

where $C_i \equiv \lambda_i^C c_0 T_i^{-1/\zeta} \bar{h}_i^{(\zeta+1)/\zeta}$ is the construction cost at the cap. In a dense city with binding plans, most parcels are in the constrained regime (Appendix Table E5). This is why the paper focuses on the extensive margin, whether a parcel is redeveloped at all, rather than on how much is built once it is.

3.4. The land market

The most a developer is willing to pay for parcel i at rent R is the value of the project net of every cost she bears, deflated by the transfer tax. I call it the developer value:

$$V_i^D(R) \equiv \frac{\pi_i(R) - F_i - \tau^A h_i(R)}{1+\tau^M}. \tag{5}$$

The gain from redeveloping, which I call the surplus, is the developer value minus the standing value the owner gives up:

$$S_i(R) \equiv V_i^D(R) - V_i^w(R). \tag{6}$$

The price of land is a transfer and does not enter the surplus.[2] The land market determines how the developer value is split, and the next section shows that the split also changes when parcels are sold.

Assumption 4 (Offers and parcel supply)**.** Municipality m has N_m developer groups indexed by j. Developer j offers owners a share $\theta_{jm} \in (0,1)$ of the developer value. Owners hold private taste shocks over keeping their parcel and over the identity of the buyer, with a two-level nested structure. The outer nest separates selling from holding, with elasticity η^S. The inner nest separates developers, with elasticity $\eta^D > \eta^S$. Measured in units of developer value, the supply of parcels to developer j is

$$L_{jm} = \left(\frac{\theta_{jm}}{\Theta_m}\right)^{\eta^D} \left(\frac{\Theta_m}{\bar{\Theta}}\right)^{\eta^S} \bar{L}_m, \qquad \Theta_m = \left(\sum_{k=1}^{N_m} \theta_{km}^{1+\eta^D}\right)^{\frac{1}{1+\eta^D}}, \tag{7}$$

where $\bar{L}_m$ is the stock of ripe parcels and $\bar{\Theta}$ the offer index at which all of them would sell. Developers choose acquisitions à la Cournot. Shares are stable over time.

The elasticity η^S measures how readily owners sell when offers rise. The elasticity η^D measures how readily they switch from one developer to another. The same structure arises from a nested logit over discrete choices, as in the labour literature (Card et al., 2018; Berger et al., 2022). The constant $\bar{\Theta}$ captures the owners' outside option. It scales how many parcels sell but does not enter the markdown. In the dynamic problem below, it is replaced by each parcel's standing value V_i^w.

Let w_{jm} denote developer j's share of land payments in municipality m, and $HHI_m = \sum_j w_{jm}^2$ the Herfindahl index of these shares.

Proposition 1 (Markdown)**.** *Developer j offers*

$$\frac{1}{\theta_{jm}} = 1 + \frac{w_{jm}}{\eta^S} + \frac{1 - w_{jm}}{\eta^D}. \tag{8}$$

The share of developer value that owners receive in municipality m, $\bar{\theta}_m \equiv \sum_j \theta_{jm} L_{jm} / \sum_j L_{jm}$, is

$$\frac{1}{\bar{\theta}_m} = 1 + \frac{HHI_m}{\eta^S} + \frac{1 - HHI_m}{\eta^D}. \tag{9}$$

Proof in Appendix B.

The offer falls with the developer's market share. A developer with a negligible share faces the elasticity η^D of owners switching between developers. A developer with the whole market faces only the elasticity

[2] I evaluate the development tax at the density chosen in (2) and neglect its feedback on that choice. In the constrained regime, $h_i(R) = \bar{h}_i$ and the treatment is exact.

η^S of owners deciding whether to sell. Aggregation is exact: the owners' share in a municipality, which I call the land share $\bar{\theta}_m$, depends only on the local Herfindahl index and the two elasticities.

Corollary 1 (Benchmarks)**.** *A single developer pays the land share $\eta^S/(1+\eta^S)$. Many small developers pay $\theta^{atom} \equiv \eta^D/(1+\eta^D) < 1$. Owners receive the full developer value, $\bar{\theta}_m = 1$, only if developers are also perfect substitutes, $\eta^D \to \infty$.*

The corollary has a direct policy implication. Deconcentration alone cannot make the land market competitive. With many small developers, owners still receive only θ^{atom} of the developer value, because each owner prefers some developers to others.

The elasticities η^S and η^D and the market shares are exogenous. The land share $\bar{\theta}_m$ follows from them by (9). I do not model why few developers operate. Concentration could reflect efficient scale or barriers to entry. Either way, given the number of developers, the markdown distorts the owners' decision, which is what the welfare analysis requires.

Remark 1 (Why owners are paid less than the developer value)**.** The markdown rests on two features of the land market. Developers do not observe what an owner would accept, and they cannot commit to pay each owner her own reservation value. With private values, no trading mechanism is both efficient and voluntary (Myerson and Satterthwaite, 1983), and a buyer facing sellers with private reservation values optimally posts an offer below her own value (Riley and Zeckhauser, 1983). Assumption 4 is the oligopsonistic counterpart of this posted offer, expressed as a share of the developer value. In the timing problem below, I follow the owner whose taste shock for keeping her parcel is zero, so that she values her structure at its market level. The taste shocks of other owners set the elasticities, and hence the share she is offered, but not her exercise rule. This is why they do not appear in Lemma 2.

3.5. Timing: the redevelopment threshold

The owner holds the option to sell. When she sells at rent R, she receives $\bar{\theta}_m V_i^D(R)$ and gives up her standing value $V_i^w(R)$. Let β be the positive root of $\frac{1}{2}\sigma^2\beta(\beta-1)+\delta\beta-\rho=0$, which governs the value of waiting. Since $\delta<\rho$, $\beta>1$.

Lemma 2 (The owner's problem)**.** *The value of a parcel to its owner is*

$$W_i(R) = V_i^w(R) + \bar{\theta}_m\, \mathcal{O}_i^\theta(R), \qquad \mathcal{O}_i^\theta(R) \equiv \sup_\tau \mathbb{E}\left[e^{-\rho\tau} S_i^\theta(R_\tau)\right], \qquad S_i^\theta(R) \equiv V_i^D(R) - \frac{V_i^w(R)}{\bar{\theta}_m}. \quad (10)$$

Proof in Appendix B.

The owner exercises the option on the private surplus S_i^θ.[3] It differs from the surplus S_i in one place: the standing value is divided by the land share $\bar{\theta}_m$. An owner who receives only a fraction of the developer value behaves as if her building were worth more than it is.

Proposition 2 (Redevelopment threshold)**.** *The owner sells, and the parcel is redeveloped, the first time the rent reaches $R_i^\star$, the smallest root of $\beta S_i^\theta(R) = R S_i^{\theta\prime}(R)$ with $S_i^\theta(R) > 0$. In the constrained regime, $S_i^\theta(R) = a_i^\theta R - K_i^\theta$ with*

$$a_i^\theta = \frac{1}{\rho - \delta}\left[q_i - \frac{h_i^{old}}{\bar{\theta}_m}\right], \qquad K_i^\theta = \frac{F_i + C_i + \tau^A \bar{h}_i}{1+\tau^M} + \frac{v_i^C}{\bar{\theta}_m \rho}, \qquad q_i \equiv \frac{(1-\tilde{t}^F)\,\lambda_i^R \bar{h}_i}{(1+\tau^V)(1+\tau^M)},$$

where q_i is the after-tax, time-discounted new floorspace. Provided $a_i^\theta > 0$,

$$R_i^\star = \frac{\beta}{\beta - 1}\,\frac{K_i^\theta}{a_i^\theta}. \tag{11}$$

In the interior regime for a parcel without standing housing, and abstracting from the development tax, $S_i^\theta(R) = b_i R^{\zeta+1} - K_i^{int,\theta}$, where b_i is the coefficient of $R^{\zeta+1}$ in the interior branch of (4) divided by $1+\tau^M$, and $K_i^{int,\theta} \equiv F_i/(1+\tau^M) + v_i^C/(\bar{\theta}_m \rho)$. Provided $\beta > \zeta + 1$,

$$R_i^\star = \left(\frac{\beta}{\beta - \zeta - 1}\right)^{\frac{1}{\zeta+1}} \left(\frac{K_i^{int,\theta}}{b_i}\right)^{\frac{1}{\zeta+1}}. \tag{12}$$

Proof in Appendix B.

The ratio K_i^θ / a_i^θ is the static break-even rent, at which the private surplus is zero. The factor $\beta/(\beta - 1) > 1$ is the premium the owner requires for giving up the option to wait. A planner facing the same irreversibility and uncertainty would require it too. In the estimation I compute $R_i^\star$ numerically for every parcel, by maximising $S_i^\theta(R) R^{-\beta}$ over R, with the development tax in both regimes. The closed forms serve as checks.

Let R_i^{SP} denote the threshold with $\bar{\theta}_m = 1$. The next section shows it is the threshold a planner would choose.

[3] An owner who sells to developer j receives $\theta_{jm} V_i^D$. Lemma 2 replaces the buyer-specific share by the municipal land share $\bar{\theta}_m$, the payment-weighted harmonic mean of the θ_{jm} in (9). The approximation error is of second order in the dispersion of market shares within the municipality.

Proposition 3 (Concentration raises the threshold)**.** *(i) The threshold $R_i^\star$ decreases with the land share $\bar{\theta}_m$, so $R_i^\star \geq R_i^{SP}$, with equality if and only if the parcel has no standing value, $V_i^w = 0$. (ii) In the constrained regime, if $\bar{\theta}_m \leq h_i^{old}/q_i < 1$, the parcel would be redeveloped by a planner but is never redeveloped privately. Proof in Appendix B.*

The markdown matters only for parcels that carry a structure. On empty land, the owner gives up nothing, so receiving a share of the developer value scales her payoff without changing when she sells. Concentration then only transfers value. On a built parcel, the markdown inflates the standing value and delays redevelopment. Where the standing housing is large relative to what could be built, it can block redevelopment for ever. Concentration is therefore a failure specific to redevelopment, not to construction on open land.

Remark 2 (Depreciation and vintage)**.** Standing structures do not depreciate, and standing floorspace earns the rent of new floorspace. Both assumptions overstate the standing value of old buildings, and both keep the problem one-dimensional. With depreciation, the standing value would fall over time relative to the developer value, and every parcel with $a_i > 0$ would eventually become ripe. The parcels that Proposition 3(ii) blocks for ever are then parcels that concentration delays beyond any horizon of interest, and their count is an upper bound.

Remark 3 (Delay due to concentration)**.** Under the perceived process, if $\delta > \sigma^2/2$, the expected time for the rent to rise from its current level to a threshold $R^\star$ is $\ln(R^\star/R)/(\delta - \sigma^2/2)$. The expected delay that concentration imposes on parcel i is therefore

$$d_i = \frac{\ln\left(R_i^\star / R_i^{\mathrm{SP}}\right)}{\delta - \sigma^2/2}, \tag{13}$$

which does not depend on the current rent. In the constrained regime, $R_i^\star/R_i^{\mathrm{SP}} = (K_i^\theta/K_i)(a_i/a_i^\theta)$, where K_i and a_i are K_i^θ and a_i^θ at $\bar{\theta}_m = 1$. This is a partial-equilibrium measure. Section 6 reports its general-equilibrium counterpart.

3.6. The planner and welfare

Definition 1 (Planner)**.** The planner faces the same technology as private parties: the same fixed cost F_i, construction lag ℓ_i, demand uncertainty, taxes, and density caps. It chooses when to redevelop each parcel to maximise the expected surplus $\mathbb{E}[e^{-\rho\tau} S_i(R_\tau)]$, which counts the developer value and the standing value at par.

The planner faces frictions: it still pays the fixed cost, waits out the construction lag, and values the option to wait. The only difference with the private equilibrium is that it does not mark down the standing value. By Proposition 2 at $\bar{\theta}_m = 1$, its threshold is R_i^{SP}. Developer rents are transfers and do not enter its objective. Taxes and caps are held fixed, and Remark 5 discusses what changes when they are not.

Proposition 4 (Welfare)**.** *Suppose taxes are zero and caps fixed. Let total surplus be the sum of owner and developer values.*

(i) The markdown is a market failure. *For every parcel with a standing value, the private threshold exceeds the planner's. Raising the land share $\bar{\theta}_m$ towards one raises total surplus to first order.*

(ii) Frictions are not. *Absent the markdown, $\bar{\theta}_m = 1$, a marginal subsidy s to the fixed cost changes total surplus by an amount of order s^2. With a shadow cost of public funds $\lambda_g > 0$, its net effect is negative. To first order in s, it equals $-\lambda_g$ times the expected outlay. The same holds for transfers that compensate the construction lag or the risk of waiting.*

(iii) A subsidy to sellers restores efficiency. *A subsidy paid to sellers at the ad valorem rate $\tau_m^{sell} = 1/\bar{\theta}_m - 1$ on the land price restores the planner's threshold for every parcel in municipality m. The rate depends only on the local Herfindahl index, η^S, and η^D. It leaves offers and developer rents per parcel unchanged.*

(iv) Fixed-cost subsidies and deconcentration do not. *A lump-sum subsidy to the fixed cost restores the planner's threshold only if it is set parcel by parcel, as a function of the fixed cost and the standing value. A uniform subsidy raises total surplus to first order on parcels with a standing value, whose threshold is too high, and lowers it to second order on parcels without one, whose threshold is already efficient. No subsidy bounded by the fixed cost can unblock a parcel blocked by concentration. As $HHI_m \to 0$, the land share tends to $\theta^{atom} < 1$, so deconcentration leaves the threshold above the planner's.*

Proof in Appendix B.

This proposition organises the paper. Subsidies to the fixed cost, density relaxations, and VAT cuts can all be justified, but not as corrections of a market failure. Their case rests on other grounds: fiscal self-financing (Section 6.1), the land-take externality of building on open land, or a failure inside the fixed cost itself such as co-ownership hold-outs. The markdown is different. Its correction raises welfare on its own, and its structure makes it cheap to target. Because the markdown is a common proportional

wedge within a municipality, the planner does not need to observe owners' taste shocks, fixed costs, or standing values. It needs only the local Herfindahl index and two elasticities.

Remark 4 (Two implementations)**.** The subsidy to sellers is costly, because it is also paid to owners who would have sold anyway. A self-financing alternative taxes the rental flow of standing structures at rate $1-\bar{\theta}_m$, with new buildings exempt. The owner's payoff from selling becomes $\bar{\theta}_m V_i^D - \bar{\theta}_m V_i^w = \bar{\theta}_m S_i$, so she sells at the planner's threshold. The tax raises revenue but falls on every owner of a standing structure. The tax on under-used land of Appendix E.3 is a targeted version of this instrument.

Remark 5 (Policy wedges)**.** Taxes are transfers. A planner who counts tax revenue as social value would redevelop on the pre-tax surplus, whereas private parties exercise on the after-tax surplus. Taxes therefore raise every private threshold above the one such a planner would choose. A subsidy that offsets them has a first-order welfare effect equal, at the margin, to the tax revenue paid by each project it induces minus the shadow cost of the funds it uses. The budget rule of Section 6.1 is the operational version of this criterion. The same applies to density caps, unless they correct an externality the model does not contain.

3.7. Identification of the fixed cost and redevelopment sample

The price of a parcel that has not been redeveloped is its value to its owner, W_i in (10). It equals the standing value plus the owner's share of the option to redevelop. Given the land share, the price therefore reveals the option, and the option reveals the fixed cost.

Proposition 5 (Identification of the fixed cost)**.** *Fix the structural parameters, the current rent R_{i0}, and the land share $\bar{\theta}_m$. The option value $\mathcal{O}_i^{\theta}(R_{i0};F)$ is continuous and strictly decreasing in the fixed cost F, from $\mathcal{O}_i^{\theta}(R_{i0};0)$ to zero. A unique $F_i \geq 0$ rationalises the observed price $\hat{p}_i^L$ if and only if*

$$V_i^w(R_{i0}) \;<\; \hat{p}_i^L \;\leq\; V_i^w(R_{i0}) + \bar{\theta}_m\, \mathcal{O}_i^{\theta}(R_{i0};0). \tag{14}$$

Proof in Appendix B.

The land share comes from sales of ripe parcels and from developer margins, and it is solved jointly with the inversion (Section 4.2). Given the land share, the fixed cost is a fully determined residual, parcel by parcel. No parameter is chosen to reproduce when parcels are redeveloped, so the permit data of Fact 1 test the model rather than fit it. Ignoring the markdown would overstate fixed costs, because part of

the gap between price and developer value would be attributed to them. Location quality, and the location component of the land price, are both measured at the municipal level. Within a municipality, the recovered fixed cost therefore varies with parcel and structure characteristics, not with location.

F_i is a property of parcel i: what redeveloping it would cost, whether or not it is ever redeveloped. A parcel priced at or below its standing value violates the lower bound of (14). Its structure is worth more than any building the market expects on it, and it would not be redeveloped at any fixed cost. The price bound, not the absence of a transaction, distinguishes it from a parcel with a high but finite fixed cost.

What the fixed cost contains F_i is a residual. It contains the physical cost of clearing the site: demolition, decontamination, and ground works. It also contains every cost of redevelopment that the model does not represent and that lowers the price a parcel commands: the unanimity required in co-ownership, the notice owed to sitting tenants, heritage constraints, the premium needed to assemble contiguous parcels, and any error in the computed developer value. The price $\hat{p}_i^L$ is predicted from a hedonic model with parcel area, floorspace, unit counts, and municipality and year effects (Appendix A.3). Within a municipality, F_i is therefore a function of these characteristics, and parcels that differ only in unobserved attributes share the same fixed cost. Appendix F relates F_i to its observable components, including the demolition costs predicted from the PRLN survey.

Remark 6 (Parcels priced above the zero-cost bound)**.** A parcel can be priced above the upper bound of (14) for three reasons. The first is measurement error in the hedonic price. The second is developer heterogeneity. If the cost of redeveloping parcel i differs across developers, the parcel is bought by the developer with the lowest cost, which can lie below the parcel's mean cost. The third is a value the model does not contain, such as an amenity or an option to assemble. I exclude these parcels rather than give them a negative cost, which would make them ripe at every rent. Among parcels priced above their standing value whose option is valuable at a zero fixed cost, 73.7% are priced above the bound. The second reason predicts that they are more frequent where more developers compete, which Section 5.1 checks against the local Herfindahl index.

I recover the fixed cost only for parcels whose price lies strictly within (14). Parcels priced at or below their standing value, parcels where the cap makes the option worthless ($a_i^\theta \leq 0$), and parcels priced above the upper bound are left out of everything that depends on the fixed cost. I solve for the fixed cost by bisection between zero and 100,000 euros per square metre of parcel area (about eighty times the construction cost c_0). A parcel at that limit is never redeveloped in the model, and I drop it. The 55,186

remaining parcels form the estimation sample.

3.8. Aggregation and margins

Redevelopment is irreversible, so the state variable for the extensive margin is the running maximum of the regional rent, $\mathcal{M}_t \equiv \max_{s \le t} R_s$. Let $\Delta_u(\mathcal{M})$ be the net quality-adjusted floorspace, $\sum_i \phi_m (h_i - h_i^{\text{old}})$, delivered by parcels of use u whose threshold satisfies $R_i^\star \le \phi_m \mathcal{M}$, each building at the density chosen at its start date. A parcel starts when the running maximum crosses its threshold and delivers ℓ_u years later. Aggregate supply and rents are

$$Q_t = \bar{Q} + \sum_u \Delta_u(\mathcal{M}_{t-\ell_u}), \qquad R_t = A_t Q_t^{-\alpha}, \tag{15}$$

where $\bar{Q}$ is the initial stock. Supply at t depends on running maxima dated $t - \ell_u$ or earlier, so it is predetermined and the equilibrium rent follows without solving a fixed point.

Proposition 6 (Concentration lowers supply at every horizon)**.** *Holding the demand path fixed, the number of parcels redeveloped by any date is weakly lower under the markdown than under the planner. Delivered floorspace is weakly lower whenever parcels build at the cap. If the running maximum of rents grows without bound, the gap converges to the floorspace of parcels blocked by concentration (Proposition 3), and to zero if no parcel is blocked. Proof in Appendix B.*

Concentration therefore delays supply everywhere and caps it where standing housing is valuable. Its effect at a finite horizon mixes both, which is why Section 6 reports the time saved alongside the gain in supply.

Supply elasticity Let $r_i \equiv R_i^\star / \phi_m$ be the regional rent at which parcel i is redeveloped, $G(r)$ the mass of parcels with $r_i \le r$, and g its density. Parcel i adds $\Delta_i \equiv \phi_m [h_i(R_i^\star) - h_i^{\text{old}}]$ to the quality-adjusted stock. Abstracting from lags, the stock at running maximum r is $Q(r) = \bar{Q} + \int \mathbb{1}\{r_i \le r\}\, \Delta_i \, di$, and the supply elasticity $\varepsilon = rQ'(r)/Q(r)$ satisfies

$$Q'(r) = \underbrace{g(r)}_{\text{extensive margin}} \times \underbrace{\mathbb{E}[\Delta_i \mid r_i = r]}_{\text{intensive margin}}. \tag{16}$$

The first factor counts the parcels that cross their threshold at r. The second is the net floorspace each delivers.

Proposition 7 (Where each wedge acts). *Parcel i starts at the first date τ_i at which the running maximum reaches its threshold r_i, and it delivers at $\tau_i + \ell_i$.*

(i) Extensive margin. *The threshold r_i increases with the fixed cost F_i, the standing value (h_i^{old} and v_i^C), the option premium $\beta/(\beta-1)$, the taxes τ^V, $\tilde{t}^F$, and τ^M, and the development tax τ^A in the constrained regime. It increases with the lag ℓ_i if $\delta < \rho/2$. It decreases with the land share $\bar{\theta}_m$ for every parcel with a standing value.*

(ii) Intensive margin. *The density built at a given rent, $h_i(R)$, depends on the lag ℓ_i, the taxes τ^V and $\tilde{t}^F$, and the cap $\bar{h}_i$. It does not depend on the fixed cost, the standing value, the option premium, the land-market taxes, or the land share.*

(iii) Delay after the start. *The time between start and delivery is the lag ℓ_i alone.*

Proof in Appendix B.

Table 2: Where each wedge acts

Wedge	Extensive margin	Intensive margin	Delay after start	Faced by planner
Fixed cost F_i	✓			✓
Option value $\beta/(\beta-1)$	✓			✓
Time to build ℓ_i	✓	✓	✓	✓
DMTO, development tax	✓			✓
VAT, property tax	✓	✓		✓
Density cap $\bar{h}_i$	✓	✓		✓
Markdown $1-\bar{\theta}_m$	✓			

Source: Proposition 7. A tick means the wedge enters the object. Every wedge that acts on the extensive margin also delays the date at which a parcel starts. The markdown acts only on parcels with a standing value (Proposition 3). Taxes and caps are faced by the planner of Definition 1 because they are held fixed (Remark 5).

Where the cap binds for most redeveloping parcels, the intensive margin is fixed by the plan, and the supply elasticity is mostly a statement about how many parcels cross their threshold. Housing supply in a built-out city is then a question of whether to build at all, rather than how much. The markdown acts on the same margin as the fixed cost. Only the planner's column separates them.

Density caps and timing A higher cap never lowers the payoff π_i. It does not follow that it brings redevelopment forward. A higher cap raises the payoff only at rents above the saturation rent $\bar{R}_i$, and by more the higher the rent. It raises the value of waiting as well as the value of building.

Proposition 8 (Relaxing the cap). *Consider a parcel with $a_i^\theta > 0$ that stays in the constrained regime. Its threshold satisfies*

$$\frac{d \ln R_i^\star}{d \ln \bar{h}_i} = \varepsilon_i^K - \varepsilon_i^a, \qquad \varepsilon_i^K \equiv \frac{(1+1/\zeta)\, C_i + \tau^A \bar{h}_i}{F_i + C_i + \tau^A \bar{h}_i + (1+\tau^M) v_i^C / (\bar{\theta}_m \rho)}, \qquad \varepsilon_i^a \equiv \frac{q_i}{q_i - h_i^{old}/\bar{\theta}_m}. \tag{17}$$

Since $\varepsilon_i^a \geq 1$, a higher cap brings redevelopment forward whenever $C_i/\zeta \leq F_i + (1+\tau^M) v_i^C/(\bar{\theta}_m \rho)$. On a parcel without standing value, it delays redevelopment if and only if $C_i/\zeta > F_i$. Proof in Appendix B.

Construction cost rises more than proportionally with height. Where it is large relative to the fixed cost, a taller allowance makes the owner wait for a rent that justifies the taller building. The standing structure works the other way. A higher cap raises the floorspace gained relative to the floorspace given up, which brings redevelopment forward. Which force dominates is an empirical matter, parcel by parcel (Section 6.2).

3.9. Misallocation across parcels

Rising demand has to be accommodated by redeveloping and densifying parcels. The planner (indexed by SP) converts parcel i when the regional rent reaches $r_i^{SP} \equiv R_i^{SP}/\phi_m$. Its allocation is therefore an order: parcels are converted in increasing order of r_i^{SP}. Following Hsieh and Klenow (2009), I separate two objects. A cost enters the planner's threshold as well as the private one. A wedge enters only the private one. Write the private threshold as $r_i = (1+\omega_i)\, r_i^{SP}$, where $\omega_i > -1$ is the wedge on parcel i. The markdown wedge is non-negative (part (ii) below). A cap wedge can be negative, since a tighter cap can bring redevelopment forward (Proposition 8).

Proposition 9 (Uniform and dispersed wedges). *(i) If the wedge is the same for every parcel, $\omega_i = \omega$, the private economy converts the same parcels as the planner, in the same order. For any quantity of new floorspace, the set of converted parcels coincides. The wedge only delays: every parcel starts when the regional rent reaches $1+\omega$ times its planner threshold. (ii) In the constrained regime, the markdown wedge is*

$$1 + \omega_i = \frac{1 + \left(\frac{1}{\bar{\theta}_m} - 1\right) \frac{v_i^C}{\rho K_i}}{1 - \left(\frac{1}{\bar{\theta}_m} - 1\right) \frac{h_i^{old}}{q_i - h_i^{old}}}, \tag{18}$$

where K_i is the intercept of the surplus S_i and q_i the after-tax, time-discounted new floorspace of Proposition 2. It is zero for parcels without standing value. It increases with the markdown $1/\bar{\theta}_m - 1$, with the non-housing

standing value relative to the cost of redeveloping, and with the standing housing relative to the new floorspace. (iii) The fixed cost, the option premium, and the construction lag enter both thresholds. They are costs, not wedges. Proof in Appendix B.

Part (i) is the land-market counterpart of the result that a tax levied at the same rate on every firm does not misallocate inputs across them. Part (ii) shows that the markdown is not uniform. It is larger where concentration is higher and where the standing building is worth more relative to what replaces it. The planner and the private economy therefore disagree not only on when to convert parcels, but on which to convert first. Because demand is over quality-adjusted floorspace, they also disagree on where housing is built whenever the wedge is correlated with location quality ϕ_m. Taxes and density caps are wedges in the same sense once the planner counts tax revenue as social value (Remark 5). Section 5.3 measures how dispersed each wedge is.

Remark 7 (Owners' attachment)**.** The planner counts the standing value at its market level. It ignores the private attachment of owners, which is what generates the markdown. Part of the measured misallocation may therefore reflect owners who genuinely value their parcel more than the market does. In this sense, the measures of Section 5.3 are upper bounds.

4. Calibration

The calibration has three steps. I first fix the structural parameters, either from the literature or from aggregate moments. I then estimate how the land share falls with local concentration from sales of ripe parcels, and set its level to the operating margins of listed developers. Finally, I recover the fixed cost parcel by parcel from the price of land (Proposition 5). No step uses the timing of redevelopment. Table 3 reports every parameter.

4.1. Structural parameters

Demand Market clearing gives $\log A_t = \log R_t + \alpha \log Q_t$. I build an annual series for the Paris area from the regional rent index (*Observatoire des Loyers*) and the quality-adjusted housing stock, $Q_t = \sum_m \phi_m Q_{mt}$, where Q_{mt} is the housing floorspace of municipality m in the *Fichiers Fonciers*. Under Assumption 1, the growth rate of the demand shifter A_t is normal, so its volatility σ is the standard deviation of $\Delta \log \hat{A}_t$. The series is short, and I set the drift by hand to $\mu = 1.5\%$. The estimated drift, the mean of $\Delta \log \hat{A}_t$ plus $\sigma^2/2$, is 2.2% (Appendix C.1). The inverse demand elasticity is $\alpha = 1.49$, the inverse of

the price elasticity of housing demand estimated by Albouy et al. (2016). Their elasticity measures how much housing a household consumes when its price changes. The model needs a different object: the response of metropolitan rents to the metropolitan stock. When rents fall, households also move into the region, so metropolitan demand is more elastic than household demand. The calibrated α is therefore an upper bound on the response of rents to new supply. Because a stronger rent response deters redevelopment, it also makes the supply gains of Section 6 lower bounds. Appendix Table E4 reports the main results at lower values of α.

Every counterfactual is solved under the same simulated paths of the demand shifter A_t. A difference between two scenarios is therefore a difference in policy, not in demand.

Discounting and the value of waiting The discount rate ρ is the developers' nominal cost of capital: the average 10-year OAT yield plus a development risk premium. I set $\rho = 6.4\%$ (Appendix C.2). The perceived drift of rents is $\delta = \rho - y$, where y is the net residential yield in Île-de-France, so that the price of a rental flow is $R/(\rho - \delta)$. I set $y = 4.4\%$. The perceived drift δ is a pricing object. It capitalises market rents into observed prices at the yield y and is not a forecast. It exceeds the realised drift of rents, $\mu - \alpha g_Q = 0.6\%$ (Table G4, Panel B). Owners and developers may price in more growth than materialises, or the yield may embed risk and liquidity premia. I tie δ to prices because the standing and developer values must reproduce land prices. Table E4 reports results for $y \pm 50$ basis points. Together with σ, these two rates determine the root β of the value of waiting, and hence the option premium $\beta/(\beta - 1)$. The calibrated root is $\beta = 3.19$, above $\zeta + 1 = 2.17$, so the interior closed form (12) applies, with a premium $[\beta/(\beta - \zeta - 1)]^{1/(\zeta+1)} = 1.7$. Most thresholds lie where the cap binds: 87% of finite thresholds are in the constrained regime. When σ is small relative to δ, β is close to ρ/δ, and the premium reflects expected rent growth more than uncertainty (Lange and Teulings, 2024). Owners discount at the same rate as developers. An owner with a lower discount rate values her structure more, which acts like the attachment of Remark 7.

Construction The curvature of construction cost, $\zeta = 1.17$, comes from Combes et al. (2021). The average construction cost, $c_0 = 1{,}186$€, comes from a separate estimation on the PRLN survey (Appendix C.3). The construction lag ℓ_u runs from the date the developer commits to completion. Because the deed recorded in DVF usually follows the permit (Section 2.1), I date commitment at the earlier of the deed and the permit authorisation, by use of origin (Appendix C.4).

Density cap and parcel variables Absent a digitised cap for every parcel, I infer $\bar{h}_i$ from the 90th percentile of building height and coverage within each cadastral section, with a floor-to-ceiling height of 2.5 metres. Figure A2 shows that most new constructions stay below it. Parcel area T_i comes from the cadastre, standing housing floorspace h_i^{old} from the *Fichiers Fonciers*, and the non-housing rental flow v_i^C from the DGFiP rental values of professional premises.

Location quality Location quality ϕ_m is the ratio of the municipal rent to the regional median. Paris counts as 20 arrondissements, so the centre, where location premia vary most, is finely divided. The land price hedonic uses municipal effects at the same level (Appendix A.3). Within a municipality, the model therefore treats parcels as identical in location and different in everything else, and the recovered fixed cost does not absorb location value. The cost of this choice is that misallocation within municipalities is not measured, so the spatial component of misallocation is likely understated. I apply the regional yield to every parcel. Yields are lower in central Paris than in the inner suburbs, so the perceived drift is understated in the centre and overstated in the suburbs. The recovered fixed cost absorbs part of this gap, which shapes its spatial pattern (Figure E2).

4.2. The land share

Owners sell their parcels to developers. When one group accounts for a large share of purchases in a municipality, owners have fewer alternative buyers and receive a smaller share of the value of redevelopment. The markdown (8) captures this force, disciplined by two objects: the slope measures how fast the land share falls with concentration, and the level measures what owners would receive if developers were atomistic. Land sales bound the slope. Developer margins set the level.

Let w_{jm} be the share of group j in the land purchases of municipality m. By (8), the land share paid by j is

$$\theta_{jm} = \frac{\theta^{\text{atom}}}{1 - s\, w_{jm}}, \qquad s \equiv -\nu\, \theta^{\text{atom}}, \qquad \nu \equiv \frac{1}{\eta^S} - \frac{1}{\eta^D}, \tag{19}$$

where $\theta^{\text{atom}} = \eta^D/(1+\eta^D)$ is the land share with atomistic developers. The slope s is negative if and only if $\eta^D > \eta^S$. By (9), concentration affects the municipal land share only through the Herfindahl index: $1/\bar{\theta}_m = (1 - s\, HHI_m)/\theta^{\text{atom}}$. Given s and θ^{atom}, the observed indices deliver the land share of every municipality.

I compute each group's share of land purchases from the universe of *Sitadel* redevelopment permits, after consolidating project-specific vehicles into their parent group. Each permit is weighted by its floorspace

times the local price of new housing. Shares are computed over a rolling three-year window. The consolidation is still incomplete, so measured Herfindahl indices are too low.

Taking logs of (9), the land share falls with the Herfindahl index at the rate $d \ln \bar{\theta}_m / dHHI_m = s/(1 - s\, HHI_m)$. I estimate this relation on 5,615 *DVF* sales of housing parcels followed by a permit on the same parcel, filed at most one year before and 3 years after the sale. At such a sale the parcel is ripe, so Lemma 2 gives its price as $\bar{\theta}_m V_i^D$, and the ratio of the price to the developer value is the land share, up to the fixed cost. On a parcel that is not ripe, the price is $V_i^w + \bar{\theta}_m \mathcal{O}_i^\theta$, and the same ratio would not identify the land share. The estimating equation is

$$\ln \frac{x_i}{g_i} = \gamma_{d(i)} + \gamma_{t(i)} + b_T \ln T_i + b\, HHI_{m(i)t(i)} + u_i, \tag{20}$$

where x_i is the price per square metre of land, g_i the developer value per square metre computed from the model with a zero fixed cost, T_i the parcel area, and γ_d and γ_t département and year effects. Concentration may respond to local land values. I instrument the index with the average index of the other municipalities of the same territory (EPT). Developer groups operate across a territory, so the instrument shifts the buyer side of the market. The exclusion restriction is that concentration elsewhere in the territory affects the price of a parcel only through this channel, given département and year effects. At the mean index $\overline{HHI}$, the model implies $s = b/(1 + b\,\overline{HHI})$.

The fixed costs absorb the level of the land share in sale prices. I therefore set θ^{atom} so that the median model-implied operating margin equals the median margin of listed French developers. Operating margins also remunerate risk, overhead, and capital. Attributing the whole margin to the markdown therefore gives an upper bound on its level. The fixed costs are recovered jointly (Appendix D.3). Both elasticities then follow in closed form:

$$\frac{1}{\eta^D} = \frac{1}{\theta^{\text{atom}}} - 1, \qquad \frac{1}{\eta^S} = \frac{1}{\eta^D} - \frac{s}{\theta^{\text{atom}}}. \tag{21}$$

The slope is imprecisely estimated: $b = -0.52$ (s.e. 0.32), which implies $s = -0.59$ with a 95% confidence interval of $[-1.65, 0.17]$ (Appendix Table D1). The data cannot reject that concentration leaves the land share unchanged. I calibrate the model at the lower end of the interval, $s = -1.65$. This is the largest markdown the data allow. Combined with developer margins, it gives $\eta^D = 969.47$ and $\eta^S = 0.61$. With atomistic developers, owners would receive $\theta^{\text{atom}} = 100\%$ of the developer value. At observed concentration, they receive 62% on average, between 41% and 78% across municipalities. The effects of

concentration reported below are therefore upper bounds: the slope bounds the part of the markdown due to concentration, and the margins bound its level. Appendix Table E4 reports the main results at the point estimate of s and at $s = 0$.

4.3. Taxes

VAT is $\tau^V = 20\%$, the standard rate on new housing, with input VAT deductible. The DMTO rate τ^M combines the departmental duty (5% for 2025–2028), the municipal tax (1.2%), and a state levy of 2.37% of the departmental duty, 6.32% in total. The development tax τ^A is the statutory value per square metre in Île-de-France (1,036 € in 2025) times the sum of the municipal, departmental, and regional rates at their standard maxima (Appendix C.5). The property tax rate t_m^F is the municipal rate where it is available, and 30% elsewhere. The base share is $\psi^{\text{new}} = 0.25$. The exemption lasts $e = 2$ years.

Table 3: Calibration parameters

Par.	Name	Source	Value
μ	Drift of the demand shifter (%)	*Fichiers Fonciers*	1.5
σ	Volatility of the demand shifter (%)	*Observatoire des Loyers*, *Fichiers Fonciers*	0.9
α	Inverse demand elasticity (upper bound)	Albouy et al. (2016)	1.49
ρ	Discount rate (%)	OAT yield + developers' risk premium	6.4
y	Net residential yield (%)	Developer report (Appendix C.2)	4.4
δ	Perceived drift of rents (%)	$\rho - y$	2.0
β	Root of the value of waiting	Fundamental quadratic	3.19
ζ	Curvature of construction cost	Combes et al. (2021)	1.17
c_0	Average construction cost	PRLN	1,186
ℓ_u	Construction lag, by use of origin (years)	DVF, *Sitadel*, *BD Topo*	3.5–4.3
ϕ_m	Location quality (P10–P90)	*Observatoire des Loyers*	0.90–1.54
s	Slope of the land share (95% lower bound)	Sales followed by a permit	-1.65
η^D	Elasticity across developers	Slope s, developer margins	969.47
η^S	Elasticity of selling	Slope s, developer margins	0.61
HHI_m	Municipal Herfindahl index (median)	*Sitadel*	0.333
$\bar{\theta}_m$	Land share, mean (%)	Equation (9)	62
θ^{atom}	Land share with many small developers (%)	$\eta^D/(1+\eta^D)$	100
τ^V	VAT on new housing	Code général des impôts	20%
τ^M	DMTO rate (%)	Code général des impôts	6.32
τ^A	Development tax (€/m^2)	Code de l'urbanisme	88
$t_m^F, \psi^{\text{new}}, e$	Property tax rate (%), base share, exemption (years)	REI, otherwise Code général des impôts	30, 0.25, 2

5. Baseline results

5.1. Validation

The calibration targets developer margins and inverts land prices. It uses no timing moment and no aggregate quantity. Validation is reasonnably good.

Timing Parcel i is predicted to start between t_0 and t_1 if the running maximum of its local rent crosses its threshold $R_i^\star$ in that window. Among parcels the model ranks as ripe, 0.9% are bought by a developer in the window, against a base rate of 1.8%.[4]

Aggregate flow On the parcels of the estimation sample, the private equilibrium delivers 0.55 million square metres of new floorspace a year, against 0.16 million recorded in *Sitadel* on the same parcels over 2014–2024. This last figure includes only permits with a recorded developer, hence understating aggregate supply on those parcels.

Supply elasticity Chapelle et al. (2023) estimate a supply elasticity of 0.35 for the Paris area. They use long differences over 1999–2019 and instrument local demand with a shift-share design, following Saiz (2010). I compute the model analogue. Two economies face the same demand draws, one with the demand shifter A_t permanently 10% higher. After 20 years, the model elasticity is the difference in the log housing stock, in square metres, divided by the difference in the log rent between the two economies. Both are equilibrium outcomes, and the demand shift plays the role of the instrument. The model implies 0.25. The main difference with the data is coverage. In the model, only idle parcels of the estimation sample can add floorspace, while the stock includes every dwelling.

5.2. Decomposing the supply shortfall

Shapley decomposition Let Q^{private} be the quality-adjusted stock at the 20-year horizon in the private equilibrium, and Q^0 the stock when every wedge is switched off, under the same demand paths. The wedges are $\mathcal{K} = \{\theta, \beta, \ell, F, \tau, \bar{h}\}$: the markdown (land share $\bar{\theta}_m \to 1$), the option value (exercise at the static break-even rent, $\beta/(\beta-1) \to 1$), time to build ($\ell_u \to 0$), the fixed cost ($F_i \to 0$), taxes (all rates set to zero), and the density cap (lifted to the 99th percentile of built floor-area ratios in the region). Switching off uncertainty alone would not remove the option value, since waiting has value whenever rents grow. For a subset $\mathcal{S} \subseteq \mathcal{K}$, let $Q(\mathcal{S})$ be the stock once every wedge in $\mathcal{S}$ is switched off. With $K = 6$

[4] Four features of the data and the model make this check demanding. Parcel characteristics are measured in 2025, so a parcel bought in the window and already rebuilt enters with its new structure, which raises its standing value and its threshold: the check is biased against the model. Location quality is measured by municipality, so the classifier cannot rank parcels by location within a municipality. The fixed cost is a function of the characteristics that enter the hedonic price, so parcels that differ only in unobserved attributes share the same threshold. And owners have no heterogeneity beyond the parcel: a parcel ripe at the current rent should be redeveloped at once. The parcels ripe at the current rent but not redeveloped (Table 5) measure what the model leaves out, such as co-ownership and tenant protection.

wedges, the Shapley value of wedge k is

$$\varphi_k = \sum_{\mathcal{S} \subseteq \mathcal{K} \setminus \{k\}} \frac{|\mathcal{S}|!\,(K - |\mathcal{S}| - 1)!}{K!} \left[Q(\mathcal{S} \cup \{k\}) - Q(\mathcal{S}) \right], \tag{22}$$

the average marginal contribution of k over every order in which the wedges could be removed. The values sum exactly to $Q^0 - Q^{\text{private}}$. I group them into the market failure (θ), frictions (β, ℓ, F), and policy wedges (τ, $\bar{h}$). 1.5% of the shortfall comes from the markdown, 68.5% from frictions, and 30.0% from taxes and density caps (Table 4). The fixed cost alone accounts for 53.5%. In the frictionless benchmark, quality-adjusted new floorspace at twenty years is 17.2 times that of the private equilibrium, so shares divide a very large gap. They do not measure what policy can recover, and they average each wedge over its interactions with the others. The last column of Table 4 reports the gain from removing each wedge alone from the private equilibrium. The fixed cost remains the largest wedge (48.9%, or 7.9 times private new supply). Taxes and caps matter much less alone (9.1% and 0.4%), so their Shapley share comes mostly from interactions with the fixed cost.

Table 4: Shapley decomposition of the supply shortfall

Wedge	Group	Shapley share of the gap (%)	Removed alone (% of the gap)
Markdown ($\bar{\theta}_m \to 1$)	Market failure	1.5	1.4
Option value	Friction	10.3	10.7
Time to build	Friction	4.7	3.7
Fixed cost	Friction	53.5	48.9
Taxes	Policy	20.0	9.1
Density cap	Policy	10.0	0.4
Market failure	Total	1.5	1.4
Frictions	Total	68.5	63.3
Policy wedges	Total	30.0	9.4

Data: model estimation. Each cell is the Shapley value φ_k of (22), as a share of $Q^0 - Q^{\text{private}}$ at the 20-year horizon, computed over the $2^6 = 64$ subsets of wedges. Group totals for the market failure, frictions, and policy wedges.

Which parcels are held back by what For every idle parcel of the estimation sample, I ask whether it would be ripe at the current rent in the frictionless benchmark. If not, it is correctly idle. If it would be, I switch the wedges off one group at a time to find what holds it back: the markdown, frictions, taxes and caps, or several of them. The markdown alone holds back 0% of the idle parcels ripe in the frictionless benchmark but not in the private equilibrium, frictions alone 76%, and taxes and caps alone 0%. The median delay that the markdown imposes on a parcel with a standing structure, from Remark 3, is 1.8 years.

The markdown does more on parcels outside the estimation sample. Across all idle parcels, 112,299 (23%) are blocked by concentration for ever (Proposition 3). Their floorspace is not in the decomposition above, so the Shapley share of the markdown is a lower bound. Their private option is worthless, so their price reveals no fixed cost, and they are not part of the idle stock on which the counterfactuals of Section 6 are computed. Remark 2 explains why their count is an upper bound.

Table 5: Which idle parcels are held back by what

Row	Parcels	Share of the idle stock (%)	Share of potential floorspace (%)	Median delay due to the markdown (y)
Correctly idle	287	0.5	0.2	8.7
Held back by markdown only	1	0.0	0.0	42.7
Held back by frictions only	38,634	72.0	83.9	0.8
Held back by taxes+caps only	0	0.0	0.0	–
Held back by several wedges	12,270	22.9	12.4	5.3
Ripe at the current rent	2,450	4.6	3.5	4.1
Memo: blocked by concentration, all idle parcels	112,299	22.7	–	–

Data: model estimation. Idle parcels of the estimation sample. Rows: correctly idle, held back by the markdown only, by frictions only, by taxes and caps only, by several wedges, ripe at the current rent. Columns: parcel count, share of the idle stock, share of the floorspace these parcels would add in the frictionless benchmark, median delay due to the markdown on parcels with a standing structure. Memo row: idle parcels blocked by concentration for ever, among all idle parcels.

5.3. Misallocation of redevelopment

Proposition 9 implies that the markdown changes which parcels are converted, not only when. I measure this in four ways. Table 6 reports each measure for the markdown, and for taxes and density caps for comparison.

Dispersion of the wedge The markdown wedge ω_i of (18) is zero on vacant parcels. On built parcels, its median is 0.04 and its interquartile range is [0.01, 0.11]. Its correlation with location quality ϕ_m is -0.17.

Order of redevelopment The rank correlation between private and planner thresholds, across parcels with a finite threshold, is 0.98. Over twenty years, 9% of new floorspace is built on parcels that the planner would not have converted to deliver the same physical quantity.

Allocation versus quantity Let Q^{private} and Q^{SP} be the quality-adjusted floorspace delivered in twenty years in the private equilibrium and by the planner. I hold the physical floorspace delivered privately fixed and reallocate it to the parcels the planner would convert first. The resulting quality-adjusted supply Q^{realloc} splits the gap into an allocation component, $Q^{\text{realloc}} - Q^{\text{private}}$, and a quantity component, $Q^{\text{SP}} - Q^{\text{realloc}}$. The allocation component is -8% of the gap. This is the land-market counterpart of the exercise of Hsieh and Klenow (2009), who hold inputs fixed and reallocate them across firms.

Resource cost The reallocation above weights floorspace by ϕ_m, whereas the planner orders parcels by surplus. It can therefore come out negative: when the wedge is larger in low-quality locations, the private economy builds more in good ones. The welfare-relevant measure holds output fixed and compares costs. For a set of parcels $\mathcal{P}$ converted at dates T_i, let

$$\mathcal{C}(\mathcal{P}) = \sum_{i \in \mathcal{P}} \mathbb{E}\left[e^{-\rho T_i} \left(F_i + C_i + V_i^w(R_{T_i}) \right) \right] \tag{23}$$

be the discounted resource cost of conversion: the fixed cost, the construction cost, and the standing value given up. Taxes are transfers and do not enter. Let $\mathcal{P}^{\text{private}}$ be the parcels converted privately by year 20, and $\mathcal{P}^{\text{SP}}$ the parcels the planner converts, in its own order, until it delivers the same quality-adjusted floorspace. The share $1 - \mathcal{C}(\mathcal{P}^{\text{SP}})/\mathcal{C}(\mathcal{P}^{\text{private}})$ is the resource cost that misallocation wastes. It is -19.2%.

Location of new housing The average location quality of new floorspace is 1.30 in the private equilibrium and 1.27 under the planner, relative to a regional median of one.

Two caveats apply. Location quality is measured by municipality, so these measures miss misallocation within municipalities. And the planner ignores owners' attachment (Remark 7), so they are upper bounds on the misallocation due to the markdown.

Table 6: Misallocation of redevelopment

Wedge	Med. wedge	IQR	Corr. with ϕ_m	Rank corr. of thresholds	Floorspace built out of order (%)	Allocation share of the gap (%)	ϕ of new floorspace, with wedge	ϕ of new floorspace, without
Markdown	0.04	[0.01, 0.11]	-0.17	0.98	9.47	-8.37	1.30	1.27
Taxes	0.39	[0.34, 0.51]	-0.25	0.98	19.82	-0.96	1.30	1.23
Density cap	-0.10	[-0.14, 0.00]	-0.02	0.97	34.58	3.52	1.30	1.30

Data: model estimation. Rows: the markdown, taxes, the density cap. Columns: median and interquartile range of the wedge (built parcels for the markdown, all parcels for taxes and the cap; a negative wedge means that the tax or the cap brings the threshold forward), correlation of the wedge with location quality ϕ_m, rank correlation between private and wedge-free thresholds, share of new floorspace built by year 20 on parcels outside the wedge-free order, allocation share of the supply gap, and mean location quality of new floorspace with and without the wedge.

5.4. Threshold distribution

Figure 4 plots, by use of origin, the distribution of the threshold $R_i^\star$ relative to the current local rent R_{i0}. It also plots, for parcels with a standing structure, the ratio $R_i^\star / R_i^{\text{SP}}$ of the private to the planner's threshold. This ratio equals one on vacant parcels, and it rises with the standing value of the parcel (Proposition 3).

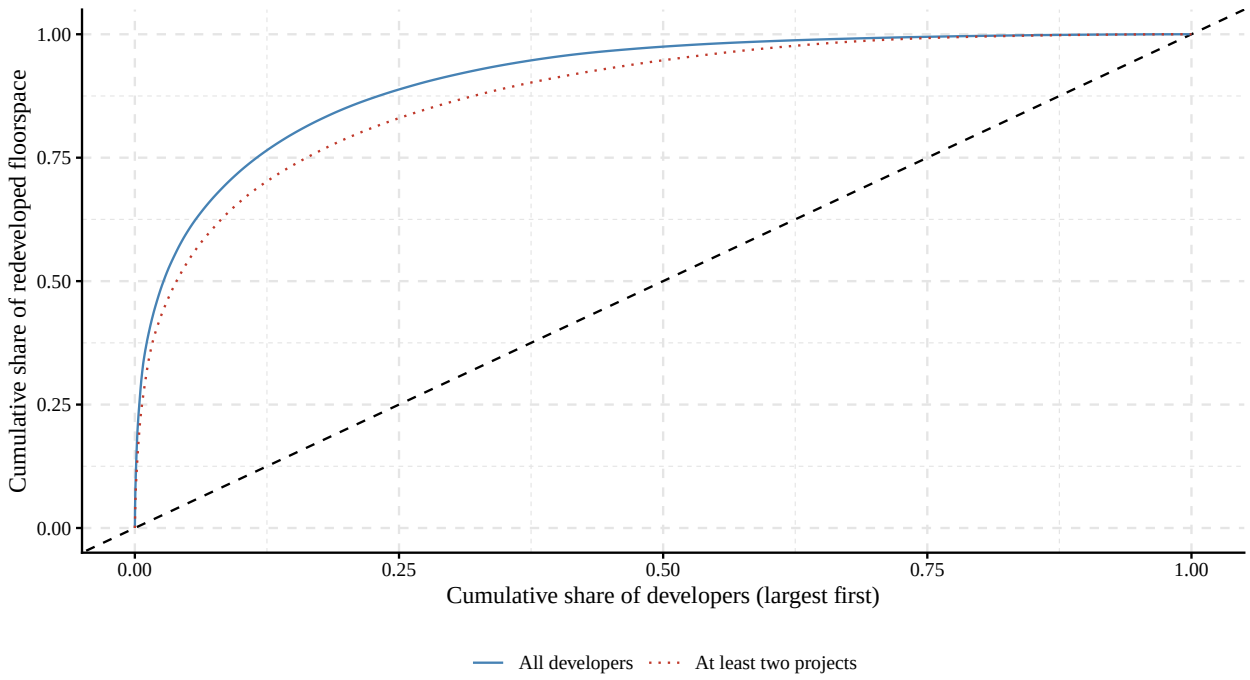


Figure 4: Redevelopment threshold relative to the current rent and to the planner's threshold

Data: model estimation. Left: distribution, by use of origin, of $R_i^\star / R_{i0}$ over the estimation sample. Right: distribution of $R_i^\star / R_i^{\text{SP}}$ over parcels with a standing structure. Log scales.

6. Counterfactual policy analysis

6.1. Outcomes

I evaluate five policies: relaxing the density cap (a policy wedge), subsidising the fixed cost (a friction), a competitive land market (the market failure), a subsidy to sellers that implements it, and an inclusionary mandate paired with a subsidy. Every counterfactual is solved under the same demand paths, on the 53,642 idle parcels of the estimation sample (Section 3.7), and reported at a 20-year horizon H. Appendix Table E1 reports the 50-year results. Every policy is reported with the same four outcomes.

Stock and rents The change in the housing stock dQ and in the market-clearing rent dR after H years, relative to the private equilibrium, in general equilibrium. The stock is the quality-adjusted stock Q_t of Assumption 1, against which rents clear, and dQ is in percent of its private-equilibrium level at the same date. Every scenario faces the same demand path, so $R_t = A_t Q_t^{-\alpha}$ gives $dR \approx -\alpha\, dQ$. I also report the additional physical floorspace delivered by H. The strength of the general-equilibrium feedback depends on α (Section 4.1). For policy (c), the stock rises by 1.45% when the running maximum of rents

is held at its private-equilibrium path.

Net cost per unit Costs and receipts are counted over the same horizon as the units. Let T_i be the start date of parcel i, s_i the subsidy it receives at that date, and r_g the government's discount rate. Let $\Phi_i(x)$ be the tax revenue a project collects in its first x years, discounted to its start date:

$$\begin{aligned}\Phi_i(x) = \tau^A h_i + \tau^M p_i^L + \mathbb{1}\{\ell_i \le x\}\, e^{-r_g \ell_i}\, \frac{\tau^V}{1+\tau^V}\, P_i^{\text{new}} \\ - \int_0^{\min(x,\, \ell_i+e)} e^{-r_g t}\, t_m^F\, b_i^{\text{old}}\, dt + \int_{\ell_i+e}^{\max(x,\, \ell_i+e)} e^{-r_g t}\, t_m^F \big(b_i^{\text{new}} - b_i^{\text{old}}\big)\, dt.\end{aligned} \tag{24}$$

The development tax and the DMTO, on the land price $p_i^L = \bar{\theta}_m V_i^D$, are paid at the start. VAT is collected at delivery on the sale price of the new units, $P_i^{\text{new}} = (1-\tilde{t}^F)\, e^{\delta \ell_i} R_{T_i} h_i / (\rho - \delta)$, which capitalises the property tax. Most new units are sold off plan (VEFA), and part of the VAT is collected before delivery, so this timing is conservative. The property tax on the standing structure, on the base $b_i^{\text{old}} = \psi^{\text{new}}(R_{T_i} h_i^{\text{old}} + v_i^C)$, is lost from the start date, when the structure is emptied. The new building pays on the base $b_i^{\text{new}} = \psi^{\text{new}} R_{T_i} h_i$ from the end of the exemption. Bases are evaluated at the rent at the start. I set $r_g = 3.0\%$. The net fiscal balance of policy π by horizon H is

$$B^\pi(H) = \sum_i \mathbb{E}\Big[e^{-r_g T_i^\pi}\, \mathbb{1}\{T_i^\pi \le H\}\, \big(\Phi_i(H - T_i^\pi) - s_i^\pi\big)\Big]. \tag{25}$$

The net cost of the policy is $NC^\pi(H) = B^0(H) - B^\pi(H)$, where B^0 is the balance without the policy. A negative net cost is a gain for the budget. The net cost per additional unit divides $NC^\pi(H)$ by the additional physical floorspace delivered by H, in units of 60 square metres, the median in the area. A project that only the policy makes possible enters at its full revenue. A project the policy merely brings forward enters only for the revenue it collects earlier. A policy is self-financing if $NC^\pi(H) \le 0$. Counting receipts until H only is conservative, since projects started late in the window collect little property tax before H. VAT accrues to the State and the other taxes to local governments, so the balance is consolidated.

Time saved Let $\bar{Q}^0(H)$ be the mean quality-adjusted stock at H without the policy, and $\bar{Q}^\pi(t)$ the mean quality-adjusted stock at date t with policy π. The time saved is

$$\Delta t^\pi \;=\; H - \min\big\{t :\; \bar{Q}^\pi(t) \;\ge\; \bar{Q}^0(H)\big\}. \tag{26}$$

If the policy reaches in twelve years the stock the baseline reaches in twenty, it saves eight years. A policy that delays supply saves a negative amount. The measure separates two effects that supply at a fixed horizon mixes: a policy can raise how much is built, and it can bring forward what would be built anyway.

Where housing is built The average location quality ϕ_m of new floorspace. Because dQ weights each new square metre by the location quality of its municipality, a policy that builds the same physical floorspace in better locations raises dQ more.

6.2. Policy (a): relax the density cap

The first policy relaxes the density limit. I evaluate two reforms. The first raises every cap to the median floor-area ratio across municipalities, in municipalities whose median cap is below it. The second lifts every cap to the 99th percentile of the floor-area ratios built in the region. Lifting the cap without bound is well posed only if $\beta > \zeta + 1$, since otherwise the value of waiting for a taller building diverges. The calibration satisfies this condition, with $\beta = 3.19$ against $\zeta + 1 = 2.17$, and Appendix Table E4 reports β at other values of ρ and y. I stop at the 99th percentile because the inferred envelope has no support in the data beyond it. The cap binds at the baseline rent for 42% of parcels. Raising it where it does not bind buys nothing.

Raising the caps to the median raises the stock by 0.51% after twenty years, lowers rents by 0.76%, and saves 1.7 years. Lifting the caps raises the stock by 0.27%, with 0.6 million square metres of additional floorspace, lowers rents by 0.40%, and saves 1.0 years. The reform costs nothing directly, and the induced projects pay taxes. Its net cost is -67.2 thousand euros per additional unit. Table 7 reports both reforms.

Proposition 8 explains why the two reforms can rank differently at different horizons. A higher cap delays the parcels where construction cost at the cap is large relative to the fixed cost. Lifting caps to the 99th percentile raises the threshold of 70% of the parcels it reaches. These parcels redevelop later but build more. Redevelopment is a single choice of density, so an owner who builds now gives up the option to build taller later. Below the cap, the surplus grows like $R^{\zeta+1}$ and the discount factor to the date the rent reaches R falls like $R^{-\beta}$. The closer β is to $\zeta + 1$, the less the value today of building at a higher rent falls with that rent, and the longer owners wait for a taller building. At a twenty-year horizon, the delay can outweigh the additional floorspace (Appendix Table E1 reports the fifty-year horizon).

Table 7: Policy (a): density relaxation

Scenario	dQ (%)	dR (%)	Net cost(m€)	Net cost/unit (k€)	Time saved	Additional floorspace(m m2)	Location quality
Density: caps raised to the median	0.51	-0.76	-1,308.74	-52.03	1.67	1.51	1.27
Density: caps lifted	0.27	-0.40	-711.88	-67.21	1.00	0.64	1.30

Data: model estimation. Columns: change in the quality-adjusted stock and in the rent after 20 years, in percent of the private equilibrium, net cost to the consolidated budget by year 20, total and per additional unit of 60 square metres (negative: a gain), years saved to reach the baseline stock of year 20, equation (26), additional physical floorspace by year 20, and mean location quality of new floorspace.

6.3. Policy (b): the self-financing subsidy to the fixed cost

The second policy pays s euros per square metre of parcel area, capped at the fixed cost, to parcels with a positive fixed cost. The outlay on parcel i is $s_i = \min(sT_i, F_i)$, and its threshold is evaluated at the fixed cost $F_i - s_i$. I choose the largest scale that is self-financing at the 20-year horizon on a grid from 900 to 1,500 euros per square metre, in steps of 50 euros (Appendix Table E2). For comparison, I also report the scale of the national *Fonds Friches* brownfield programme, 50 euros per square metre.

The self-financing scale is 1,050 euros per square metre. It raises the stock by 3.80% and lowers rents by 5.41% after twenty years, and it saves 16.2 years. Its net cost is -0.3 thousand euros per additional unit. At the *Fonds Friches* scale, the stock rises by 0.22% and 0.8 years are saved. 42% of the outlay goes to parcels that would have started within the horizon anyway, on the same demand path, and 74% of targeted parcels receive an outlay larger than the taxes they bring in. On parcels with a standing structure, the subsidy also offsets part of the markdown, which gives it a first-order welfare gain there. It does so bluntly, because the same subsidy goes to every parcel whatever its standing value, and it cannot unblock parcels blocked by concentration (Proposition 4).

Two caveats apply. The subsidy is self-financing for the consolidated budget. Local governments, which would plausibly fund it, recover 29% of the outlay, because VAT, the largest induced receipt, accrues to the State. And the subsidy offsets a cost recovered as a residual. If part of F_i is a hold-out rent rather than a resource cost, a subsidy raises the rent instead of the supply.

Table 8: Policy (b): fixed-cost subsidy

Scenario	dQ (%)	dR (%)	Net cost(m€)	Net cost/unit (k€)	Time saved	Additional floorspace(m m2)	Location quality
Fonds Friches scale (50 euros/m2)	0.22	-0.33	-252.89	-23.77	0.75	0.64	1.29
Largest self-financing scale (1050 euros/m2)	3.80	-5.41	-44.80	-0.26	16.25	10.37	1.20

Data: model estimation. Same columns as Table 7. The first row is the scale of the *Fonds Friches* programme. The second is the largest scale on the grid that is self-financing by year 20.

6.4. Policy (c): a competitive land market

The third policy removes the market failure. Owners receive the full developer value, $\bar{\theta}_m = 1$ in every municipality, and every other wedge stays at its private value. This is the planner of Definition 1. It is a benchmark, not an instrument, and it uses no public money. Section 6.5 shows how to implement it.

The quality-adjusted stock is 1.09% higher after twenty years, with 2.9 million square metres of additional floorspace. Rents are 1.60% lower. The stock the baseline reaches in twenty years is reached 3.8 years earlier. At fifty years the stock is 0.68% higher. On the idle stock, concentration mostly delays supply, so this gain shrinks with the horizon (Proposition 6). These figures leave out the 112,299 parcels blocked by concentration for ever (Section 5.2): their fixed cost cannot be recovered, and a competitive land market would add their floorspace on top. A competitive land market also changes where housing is built. The average location quality of new floorspace moves from 1.30 to 1.27.

What deconcentration can achieve Breaking up developers lowers the Herfindahl index but leaves owners' preferences over developers in place. I set $HHI_m \to 0$ in every municipality, so that the land share rises to $\theta^{\text{atom}} = 100\%$. The stock rises by 1.08% and 3.8 years are saved, which is 100% of the gain of a competitive land market. Antitrust alone cannot restore efficiency (Proposition 4). Figure E1 shows the supply paths, and Figure H4 the mean path of every policy.

Table 9: Policy (c): a competitive land market

Scenario	dQ (%)	dR (%)	Net cost(m€)	Net cost/unit (k€)	Time saved	Additional floorspace(m m2)	Location quality
Competitive land market	1.09	-1.60	-3,733.98	-78.53	3.75	2.85	1.27
Deconcentration (HHI to zero)	1.08	-1.60	-3,725.76	-78.59	3.75	2.84	1.27

Data: model estimation. Same columns as Table 7. First row: land share set to one in every municipality. Second row: Herfindahl index set to zero, land share θ^{atom}.

6.5. Policy (d): a subsidy to sellers

The fourth policy implements the competitive land market with public money. Every owner who sells a parcel for redevelopment in municipality m receives a subsidy at the ad valorem rate $\tau_m^{\text{sell}} = 1/\bar{\theta}_m - 1$ on the land price. By Proposition 4, this restores the planner's threshold on every parcel. The rate depends only on the local Herfindahl index and the two estimated elasticities. It averages 69% across municipalities. The outlay on parcel i is $\tau_m^{\text{sell}} \bar{\theta}_m V_i^D = (1 - \bar{\theta}_m) V_i^D$: the State buys back the developer's rent on every parcel sold.

The supply path is that of policy (c): the stock rises by 1.09% and 3.8 years are saved. The difference is

the cost. The subsidy is paid to every seller, including the 83% of outlays that go to parcels that would have sold within the horizon anyway, and to owners of vacant land whose timing it does not change. Its net cost is 99.9 thousand euros per additional unit. Restricting the subsidy to parcels with a standing structure, which is observable, removes the outlay on vacant land without changing supply, since the markdown does not affect when vacant land is sold (Proposition 3). The net cost per additional unit is then 99.9 thousand euros. Table 10 reports both versions. Remark 4 describes a self-financing alternative, a tax on the rental flow of standing structures at rate $1 - \bar{\theta}_m$. Its rate is a large share of the rental flow, and it falls on owner-occupiers as well as landlords. The tax on under-used land of Appendix E.3 is a smaller, targeted version.

Table 10: Policy (d): a subsidy to sellers

Scenario	dQ (%)	dR (%)	Net cost(m€)	Net cost/unit (k€)	Time saved	Additional floorspace(m m2)	Location quality	Inframarginal (%)	Change in owners' value (m€)
Subsidy to all sellers	1.09	-1.60	4,747.97	99.86	3.75	2.85	1.27	83.07	19,088.94
Subsidy to sellers of built parcels	1.09	-1.60	4,747.75	99.85	3.75	2.85	1.27	83.07	19,088.67

Data: model estimation. Same columns as Table 7, plus the share of the outlay paid to parcels that start by year 20 without the policy, and the change in the value of parcels to their owners. Rows: subsidy to all sellers at rate $1/\bar{\theta}_m - 1$, subsidy restricted to parcels with a standing structure.

6.6. Policy (e): an inclusionary mandate with an offsetting subsidy

Since 2000, municipalities short of their target social-housing share under the *loi SRU* must write inclusionary requirements into their plan. 45% of municipalities in the sample are currently in deficit. Lebret et al. (2026) show that an affordability mandate in New York crowds out market-rate units, and that who bears the cost depends on the incentive paired with it. I evaluate the closest instrument in the French setting: a share $\omega = 25\%$ of new floorspace delivered as social housing at a rent capped at $\varrho = 50\%$ of the market rent, paired with a subsidy to the fixed cost. The project payoff becomes

$$\pi_i^{\text{cs}}(R) = \max_{h \le \bar{h}_i} \frac{1 - \tilde{t}^F}{1 + \tau^V} \lambda_i^R \frac{[(1-\omega) + \omega\varrho]\, R\, h}{\rho - \delta} - \lambda_i^C c_0 T_i^{-1/\zeta} h^{1+1/\zeta}, \tag{27}$$

and the fixed cost net of the subsidy is $F_i - s_i$. The mandate acts as a proportional tax on the value of the building at rate $\tau^{\text{IZ}} = \omega(1-\varrho)$. For a parcel without standing value in the constrained regime, it scales the slope a_i^θ of the private surplus by $1 - \tau^{\text{IZ}}$ and leaves its intercept K_i^θ unchanged, so it raises the threshold by the factor $1/(1-\tau^{\text{IZ}})$. The subsidy that offsets the mandate exactly solves $K_i^\theta - s_i/(1+\tau^M) = (1-\tau^{\text{IZ}})K_i^\theta$:

$$s_i = (1 + \tau^M)\, \tau^{\text{IZ}} K_i^\theta. \tag{28}$$

For every parcel I solve numerically for the smallest subsidy in $[0, F_i]$ that restores the threshold without the mandate. Where no such subsidy exists, I set $s_i = 0$ and the mandate binds.

The mandate alone lowers the quality-adjusted stock by 2.24% after twenty years. Paired with the offsetting subsidy, the stock changes by -0.69%, and the policy delivers 2,393 thousand square metres of social floorspace, at an implicit cost of 1.72 square metres of private floorspace forgone per square metre of social floorspace. The outlay is 1,950 million euros by year 20. For 15% of parcels, the mandate cannot be offset by a subsidy bounded by the fixed cost (Table 11).

Table 11: Policy (e): the mandate, the subsidy, and the two combined

Policy	dQ (%)	dR (%)	Social floorspace (thousand m2)	Private m2 forgone/social m2	Outlay (m€)	Receipts / outlay (%)
Mandate only	-2.24	3.45	1,400.90	5.07	0.00	–
Subsidy only	2.31	-3.36	0.00	–	3,752.97	164.27
Mandate and offsetting subsidy	-0.69	1.04	2,392.63	1.72	1,949.57	-151.35

Data: model estimation. Rows: mandate only ($\omega = 25\%$, $s_i = 0$), subsidy only (the offsetting subsidies, without the mandate), and the combined policy. Columns: changes in the quality-adjusted stock and the rent at 20 years, social floorspace delivered, private floorspace forgone per square metre of social floorspace, outlay by year 20, and induced change in tax receipts as a share of the outlay.

6.7. Taking stock

Table 12 puts the policies side by side. Figure 5 plots the net cost per additional unit against the gain in supply, and Figure 6 plots the time saved. Three lessons follow.

First, the policies act on different parcels, and therefore in different places. The density cap acts where the cap binds, and it can delay parcels where construction costs dominate (Proposition 8). The fixed-cost subsidy acts where fixed costs are high. Correcting the markdown acts on parcels with a valuable standing structure in concentrated municipalities, and not at all on empty land. It removes a dispersed wedge without creating another. Taxes and caps are dispersed wedges too (Table 6), so relaxing them also changes the order in which parcels are converted. Second, only correcting the markdown removes a market failure (Proposition 4). Relaxing caps and cutting taxes remove policy wedges, whose case rests on what the induced supply brings to the public budget (Remark 5). A subsidy to the fixed cost has to be justified the same way, unless the fixed cost itself contains a failure. Third, correcting the markdown is cheap to target but not cheap to finance. Its rate requires only the local Herfindahl index, but a subsidy reaches every seller. Deconcentration costs nothing but achieves only part of the gain.

To compare the policies on a welfare metric, I also report the marginal value of public funds (Hendren and Sprung-Keyser, 2020), MVPF = WTP/net cost: the willingness to pay of those affected divided by the net cost to the consolidated budget. A policy that costs the budget has a finite MVPF. A policy that raises the welfare of those affected without costing the budget has an infinite MVPF. In the narrow

version, those affected are owners and developers, whose willingness to pay is the change in the value of their parcels and projects. The broad version adds renters. A lower rent on the existing stock is a transfer from landlords to renters, so the broad version counts only the surplus on the additional housing, $\frac{1}{2}\, dQ_t\, dR_t$ in present value. Both versions weight every euro equally. Owners of parcels are on average richer than renters, so welfare weights that decline with income would lower the MVPF of the subsidy to sellers, whose willingness to pay is mostly a windfall to owners who would have sold anyway.

Table 12: The policies side by side

Scenario	dQ (%)	dR (%)	Net cost/unit (k€)	Time saved	Location quality	MVPF, narrow	MVPF, broad
Density: caps raised to the median	0.51	-0.76	-52.03	1.67	1.27	∞	∞
Density: caps lifted	0.27	-0.40	-67.21	1.00	1.30	∞	∞
Fonds Friches scale (50 euros/m2)	0.22	-0.33	-23.77	0.75	1.29	∞	∞
Largest self-financing scale (1050 euros/m2)	3.80	-5.41	-0.26	16.25	1.20	∞	∞
Competitive land market	1.09	-1.60	-78.53	3.75	1.27	–	–
Deconcentration (HHI to zero)	1.08	-1.60	-78.59	3.75	1.27	–	–
Subsidy to sellers	1.09	-1.60	99.86	3.75	1.27	4.22	4.25
Mandate and offsetting subsidy	-0.69	1.04	–	-2.75	1.31	-2.04	-2.03

Data: model estimation. Change in the quality-adjusted stock and in the rent after 20 years, net cost by year 20 per additional unit of 60 square metres (thousand euros, negative is a gain), years saved, mean location quality of new floorspace, narrow and broad MVPF. The competitive land market and deconcentration are benchmarks without an MVPF.

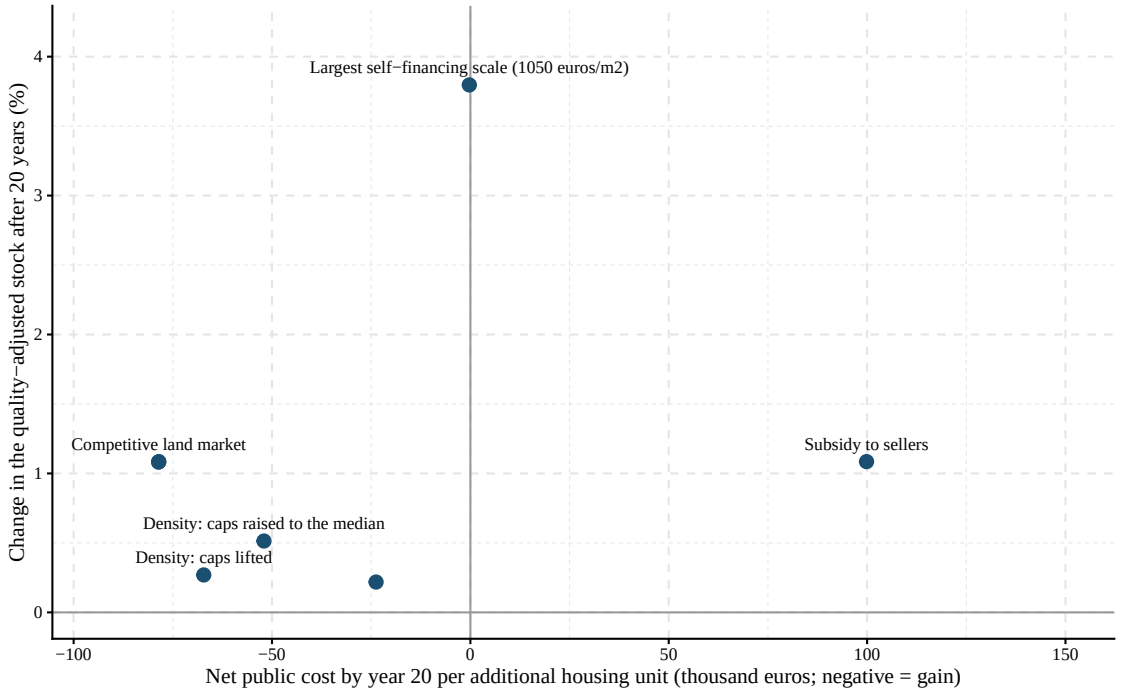


Figure 5: Supply gained and net cost per additional unit

Data: model estimation. Each point is a policy. Horizontal axis: net cost to the consolidated budget by year 20 per additional housing unit of 60 square metres, in thousand euros. A negative value is a gain for the budget. Vertical axis: change in the quality-adjusted stock after 20 years. Policies that lower the physical floorspace delivered by year 20 have no cost per additional unit and are not shown.

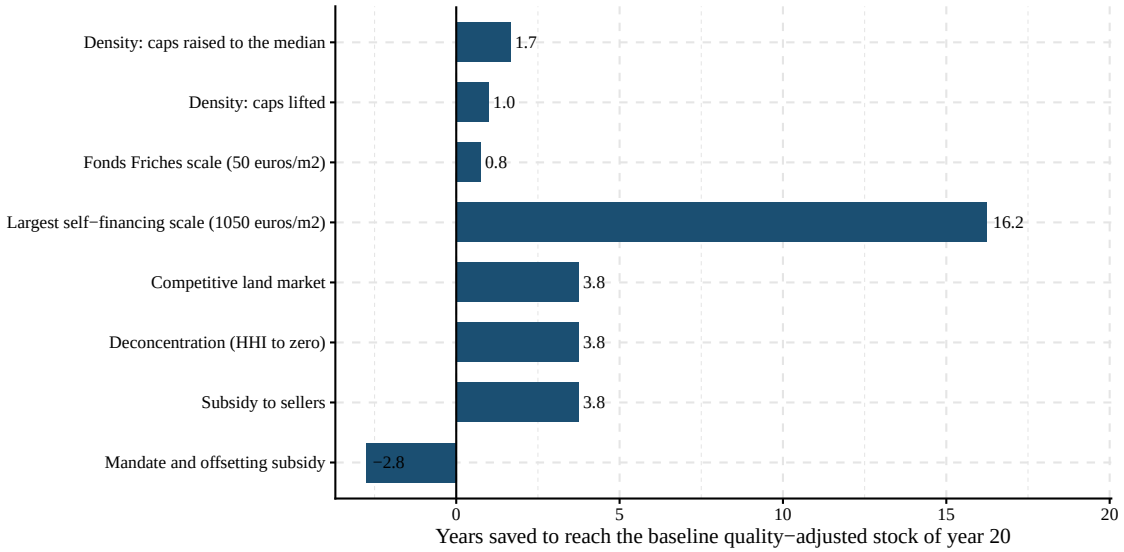


Figure 6: Time saved by each policy

Data: model estimation. Years saved to reach the quality-adjusted stock that the baseline reaches in 20 years, equation (26).

7. Conclusion

Cities that most need housing build most of it by replacing what already stands. This paper measures what holds that margin back. The price of land reveals, parcel by parcel, the cost of clearing a site, net of the delay, the taxes, and the developer's markdown. This hidden cost is the largest wedge. It accounts for 53.5% of the twenty-year supply shortfall. Taxes and density caps account for 30.0%, and the markdown for 1.5%.

The wedges differ in kind. The fixed cost, the construction lag, and the value of waiting would constrain a planner too. Taxes and caps are policy choices. They vary across parcels, so they change which parcels are converted as well as when. The markdown is a market failure. Developers pay owners less than their land is worth, and owners of built parcels sell later than a planner would let them. Some never sell. Its correction raises welfare on its own, and its rate needs only the local Herfindahl index.

The policies buy different things. Lifting the density cap raises the quality-adjusted housing stock by 0.27% after twenty years, although taller allowances also make waiting more valuable where construction costs dominate. A self-financing subsidy to the fixed cost raises it by 3.80%. It pays for itself through VAT receipts that accrue to the State rather than to the local governments that would fund it. A competitive land market raises it by 1.09% and delivers the housing of the next twenty years 3.8 years earlier. Breaking up developers cannot achieve it, because owners would still be paid less than their land is worth. A subsidy to sellers can, at a cost driven by owners who would have sold anyway. A tax on

standing structures would achieve the same at no cost to the budget, at the expense of their owners (Remark 4).

What the fixed cost contains decides which of these policies is worth its cost. If it is technological, subsidising it is a transfer that the budget can recover. If it reflects co-ownership hold-outs, tenant protection, or assembly premia, policy should act on those frictions directly.

The distinction travels beyond Paris. Any market where redevelopment, rather than construction on open land, clears rising demand faces the same choice between subsidising a friction and correcting a failure. This paper offers a way to measure, parcel by parcel and euro by euro, which one is binding, and to measure not only how much a policy builds but how much sooner it builds it.

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APPENDIX

A. Data and cleaning procedure

A.1. Parcel definitions and the fiscal-versus-physical match

The unit of analysis is the cadastral parcel, the smallest continuous unit of land with homogeneous property rights recorded by the French land registry (*cadastre*), administered by the DGFiP (*Direction générale des finances publiques*). The *Fichiers Fonciers*, the fiscal record of every taxed built unit, are redistributed for research by Cerema. They report each unit's destination, floorspace, and construction date, but they map a unit to a parcel through its postal address. This fails whenever a mailbox sits on a small contiguous parcel or several contiguous parcels share a front door. The *BD Topo*, the three-dimensional building database produced by IGN (*Institut national de l'information géographique et forestière*), gives an independent physical envelope with no address-matching problem, but no destination or ownership information.

A.2. *Sitadel*: permits and destination imputation

Sitadel is the national register of authorised building permits, maintained by the SDES (*Service des données et études statistiques*) of the ministry responsible for housing. It reports authorised floorspace and unit counts, the applicant's identity, and the permit date, but not reliably the destination of the parcel before the permit. I impute it from the last *Fichiers Fonciers* vintage preceding the permit.

A.3. DVF: transactions and the hedonic price

Demande de valeurs foncières (DVF) is published by DGFiP as open data. It records the universe of registered property transactions, with the parcel identifier, the price, and the date. DVF is used twice. The hedonic model below gives the price $\hat{p}_i^L$ of every parcel, which identifies the fixed cost. The sales followed by a permit on the same parcel identify the slope of the markdown (Appendix D).

I restrict the sample to arm's-length sales (*ventes* and *ventes de terrain à bâtir*) in Paris and the three inner-ring départements between 2014 and 2025. I exclude off-plan sales (VEFA), which price a future dwelling rather than an existing parcel. I keep transactions in which the land itself changes hands, as opposed to sales of units within a condominium. I keep single-parcel transactions, so that each price maps into one cadastral parcel. These restrictions select transactions that resemble a developer's acquisition: the buyer acquires a whole parcel with any existing structure. Multi-family buildings held in co-ownership rarely sell whole, so their predicted price extrapolates from whole-parcel sales of smaller buildings. I trim the bottom and top percentiles of the price per square metre of land, separately for vacant and built parcels. The final sample contains 105555 sales of built parcels and 9076 sales of vacant parcels.

The regressors are characteristics observed both in DVF and in the *Fichiers Fonciers*: land area S_i, total floorspace F_i^{tot}, and the number of units of each type N_i^k, with $k \in \{\text{apartments}, \text{houses}, \text{commercial}, \text{other}\}$. DVF does not report floorspace by type, so the composition of the building enters through unit counts. For a built parcel i in municipality m sold in year t, the conditional mean of the total price P_{it} is

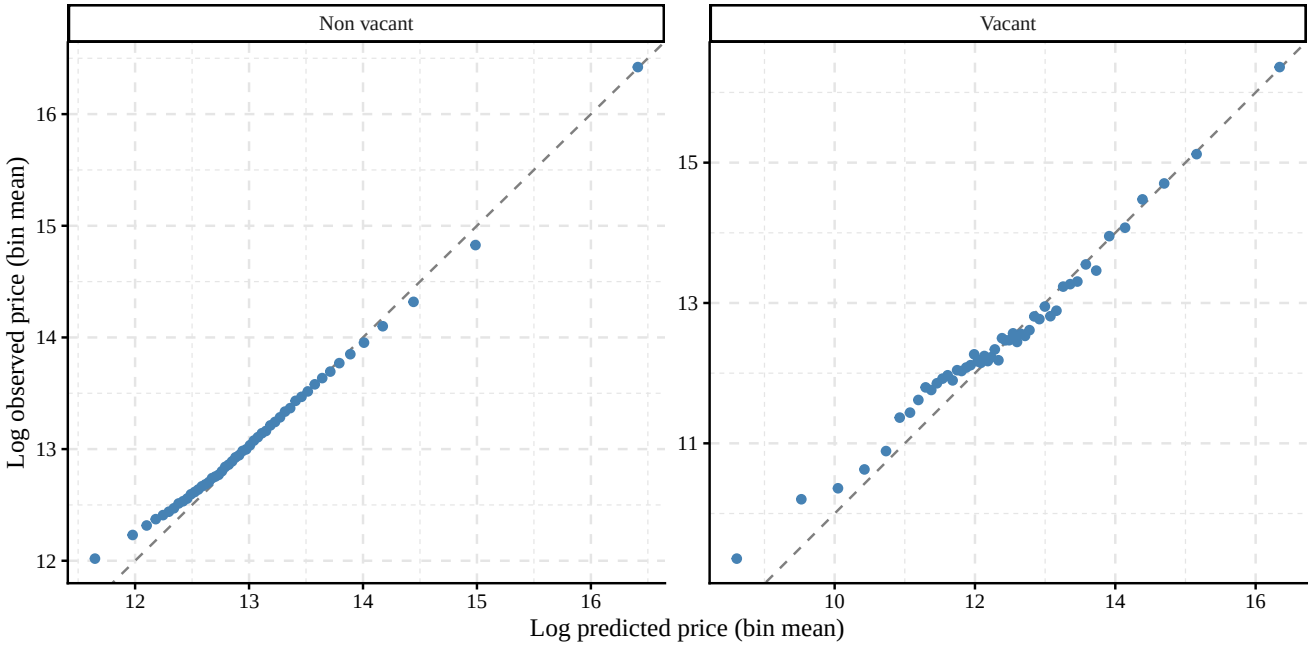
$$\mathbb{E}\left[P_{it} \mid S_i, F_i^{\text{tot}}, \mathbf{N}_i\right] = \exp\left(\gamma_m + \gamma_t + b_S \ln S_i + b_F \ln F_i^{\text{tot}} + \sum_k b_k \ln\left(1 + N_i^k\right)\right), \tag{A1}$$

where γ_m and γ_t are municipality and year fixed effects. Vacant parcels have no floorspace and no units, so I estimate a separate model with $\ln S_i$ as the only regressor. Both models target the total price in levels, the object that enters the owner's value. I estimate them by Poisson pseudo-maximum likelihood (PPML). OLS on log prices would require a retransformation to recover prices in levels, which is biased whenever the variance of the error depends on the regressors. PPML only requires the conditional mean

to be correctly specified. With municipality and year fixed effects, its first-order conditions also imply that predicted prices sum to observed prices within each municipality and year. Standard errors are clustered by municipality. Municipality effects absorb the land share $\bar{\theta}_m$ and the location quality ϕ_m, which are both measured at the municipal level.

I score every parcel of the *Fichiers Fonciers* at the 2023 year effect. I then bring the predicted price to the year of the rents, 2024, with the regional transaction-based price index. The predicted price $\hat{p}_i^L$ is the price of the whole parcel, inclusive of any structure. Figure A1 compares predicted and observed prices.[5]

Figure A1: Predicted and observed transaction prices



Notes: Binned scatter plot of observed against predicted total transaction prices, in logs. Within each panel, transactions are grouped into 50 equal-sized bins of predicted price. Each dot plots the log of the mean observed price against the log of the mean predicted price in the bin. Predictions are in-sample, from equation (A1) estimated by PPML with municipality and year fixed effects. The dashed line is the 45-degree line. Left panel: built parcels. Right panel: vacant parcels. Source: DVF, 2014–2025, départements 75, 92, 93 and 94.

A.4. The density limit $\bar{h}_i$

No complete, parcel-level digitisation of every municipality's coverage and height limits exists for the sample period. The cap $\bar{h}_i$ is therefore inferred: the 90th percentile of built coverage and height within each cadastral section, with a floor-to-ceiling height of 2.5 metres, checked against realised construction in Figure A2.

A.5. The PRLN cost survey

Prix de revient des logements neufs (PRLN) is a stratified survey of developers, drawn from the *Sitadel* permit register, that asks for realised project costs separately by construction and demolition component.

[5]Out-of-sample performance is assessed by five-fold cross-validation, with folds drawn at the parcel level so that repeat sales never straddle the training and validation samples. Planned robustness checks re-estimate the model by OLS with a smearing retransformation, vary the specification, bootstrap the predictions within municipality, and replace the parametric specification by a Poisson lasso.

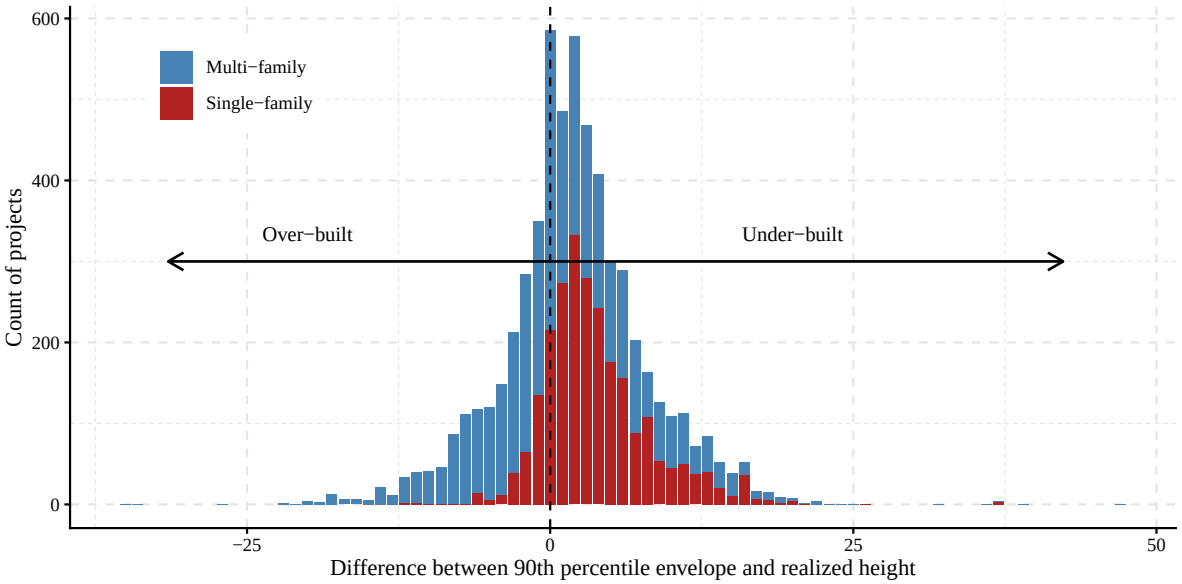


Figure A2: New construction against the inferred density limit

Data: *Fichiers Fonciers*, *Sitadel*, *BD Topo*. Distribution, across new projects, of the difference between the inferred envelope (90th percentile of building height in the cadastral section) and the realised height of the new building, for multi-family and single-family projects. Positive values are projects built below the inferred cap.

A.6. Rents, zoning, and other sources

Rents come from the *Observatoire des Loyers*, a municipal hedonic rent index, which gives the location quality ϕ_m. Land-use zoning comes from Apur (*Atelier parisien d'urbanisme*), population and income from INSEE's *Fidéli*, and ground condition from the BRGM geological survey.

A.7. Developer identifiers and concentration

Developers are identified by SIREN, the identifier INSEE's business register (*Sirene*) assigns to every legal entity, matched to *Sitadel* applicants. Many developers incorporate a project-specific company (often an SCCV) for each operation, so shares computed at the SIREN level understate concentration. I consolidate these vehicles into their parent group where the parent can be identified. Under-consolidation biases the Herfindahl index down, the land share up, and the estimated cost of concentration towards zero.

The municipal share of group j in year t is its share of land payments in municipality m over the three years up to t, where each permit is weighted by its floorspace times the local price of new housing: w_{jmt}. The municipal Herfindahl index is $HHI_{mt} = \sum_j w_{jmt}^2$. The region-wide index of Fact 5 uses floorspace shares over 2013–2024, and N^{tot} counts consolidated developers with at least one private redevelopment permit over the same window.

A.8. Donut difference-in-differences

The sample of projects is every permit delivering at least 5 housing units between 2014 and 2022, with no other project of that size within 600 metres over the event window. For each project p, delivered in year T_p, I collect every DVF resale of an existing dwelling within 600 metres of its centroid between $T_p - 5$ and $T_p + 5$. Transactions within 20 metres are dropped, since they are exposed to construction nuisance and may include the project's own units. The inner ring runs from 20 to 100 metres. Transactions between 100 and 150 metres form a buffer and are dropped. The outer ring, the comparison group, runs from 150

to 500 metres, as in Asquith et al. (2023). I stack the project samples and estimate

$$\log p_{jpt} = \alpha_{p,r(j)} + \gamma_{p,t} + \sum_{k=-5,k\neq-1}^{5} \xi_k \, \mathbb{1}\{t - T_p = k\} \, \mathbb{1}\{r(j) = \text{inner}\} + X_j' b + \varepsilon_{jpt}, \tag{A2}$$

where p_{jpt} is the price per square metre of transaction j in stack p and year t, $\alpha_{p,r}$ are project-by-ring fixed effects, $\gamma_{p,t}$ project-by-year fixed effects, and X_j hedonic controls (floorspace, number of rooms, dwelling type, and construction period). The event-time coefficients ξ_k measure the change in the price gap between the two rings relative to the year before delivery. Standard errors are clustered by project. Stacking by project compares each treated ring only with its own control ring, which avoids comparisons across cohorts that bias two-way fixed-effects estimators (Cengiz et al., 2019). The pre-period coefficients test for parallel trends. The pooled estimate of Fact 4 replaces the event-time dummies by a single post-delivery indicator.

Table A1: Donut difference-in-differences around new projects

	Project size (dwellings created)			
	All	≥ 10	≥ 50	> 100
Treated $\times$ Post	-0.073	-0.005	0.076	-0.004
	(0.058)	(0.025)	(0.057)	(0.102)
Projects	6292	1058	289	90
Transactions	37105	33868	21142	7503

Data: DVF, *Sitadel*, *BD Topo*. Treated ring 20–100 m, buffer 100–150 m (excluded), control ring 150–500 m from the project centroid. Estimates of (A2), pooled post-delivery effect. Columns: all projects (at least 5 dwellings), projects creating at least 10, at least 50, and more than 100 dwellings. Standard errors clustered by project in parentheses. Number of projects and transactions reported below each column.

Table A2: Sample construction: counts and attrition

Step	N
Parcels in the base data	552,302
With complete data and a positive density cap	502,957
Priced above the standing value	254,907
... and with a positive option value at a zero fixed cost	210,928
... and priced at or below that option value	55,557
... and a fixed cost below 100,000 euros per m2: estimation sample	55,186

Parcel count at each restriction, at the calibrated parameters and land shares: parcels in the base data, with complete data and a positive density cap, priced above their standing value (lower bound of (14)), with a positive option value at a zero fixed cost, priced at or below that option value (upper bound of (14)), and with a fixed cost inside the bracket of the bisection. The last row is the estimation sample of 55,186 parcels (Section 3.7).

Table A3: SIREN-developer matching and consolidation

Statistic	SIREN level	Parent group
Permits matched to a SIREN (%)	54.456	54.456
Floorspace of permits matched to a SIREN (%)	78.300	78.300
DVF land sales matched to a later permit (%)	2.954	2.954
Floorspace of parent groups with several SIREN (%)	0.000	31.372
Region-wide HHI	0.005	0.007
Median municipal HHI	0.333	0.333
Top-7 share (%)	13.654	16.078
N^{tot}	4,421.000	3,977.000

Match rate of *Sitadel* permits and DVF transactions to a SIREN, share of floorspace re-attributed from a project-specific vehicle to its parent group, region-wide and median municipal Herfindahl index, top-7 share, and N^{tot} before and after consolidation.

B. Proofs

Lemma 1 Using $\lambda_i^R = \kappa_i \lambda_i^C$, write (2) as

$$\pi_i(R) = \lambda_i^C \max_{h \le \bar{h}_i} \left\{ \frac{1-\tilde{t}^F}{1+\tau^V} \kappa_i \frac{R\,h}{\rho-\delta} - c_0 T_i^{-1/\zeta} h^{1+1/\zeta} \right\}.$$

The bracket is the payoff without lag evaluated at the rent $\kappa_i R$, and the constraint set depends neither on R nor on ℓ_i. Hence $\pi_i(R) = \lambda_i^C \pi_i^{0,\tau}(\kappa_i R)$ in both regimes. □

Derivation of (3) and (4) Write $X \equiv \hat{\kappa}_i \lambda_i^C R/(\rho-\delta)$ for the value of a square metre and $Y \equiv \lambda_i^C c_0 T_i^{-1/\zeta}$, so that the objective is $Xh - Yh^{1+1/\zeta}$. The interior first-order condition is $X = Y(1+1/\zeta)h^{1/\zeta}$, which gives $h_i^\star(R)$ in (3). The saturation rent solves $h_i^\star(\bar{R}_i) = \bar{h}_i$. At the interior optimum, $Y(h_i^\star)^{1+1/\zeta} = Xh_i^\star\,\zeta/(\zeta+1)$, so $\pi_i = Xh_i^\star/(\zeta+1) = (Y/\zeta)\,[\zeta X/((\zeta+1)Y)]^{\zeta+1}$, which is the interior branch of (4). Substituting $h = \bar{h}_i$ gives the constrained branch. The two branches and their derivatives coincide at $\bar{R}_i$. □

Proposition 1 Define the quantity index $\mathcal{L}_m = (\sum_k L_{km}^{(1+\eta^D)/\eta^D})^{\eta^D/(1+\eta^D)}$. Substituting (7) gives $\mathcal{L}_m = (\Theta_m/\bar{\Theta})^{\eta^S} \bar{L}_m$, and inverting the supply system gives

$$\theta_{jm} = \bar{\Theta} \left(\frac{L_{jm}}{\mathcal{L}_m} \right)^{\frac{1}{\eta^D}} \left(\frac{\mathcal{L}_m}{\bar{L}_m} \right)^{\frac{1}{\eta^S}}.$$

Taking rivals' acquisitions as given, $\partial \ln \theta_{jm}/\partial \ln L_{jm} = 1/\eta^D + (1/\eta^S - 1/\eta^D)\,\partial \ln \mathcal{L}_m/\partial \ln L_{jm}$. The last derivative is $L_{jm}^{(1+\eta^D)/\eta^D} / \sum_k L_{km}^{(1+\eta^D)/\eta^D}$. Since $\theta_{jm} L_{jm} \propto L_{jm}^{(1+\eta^D)/\eta^D}$, it equals the payment share w_{jm}. Developer j earns $(1-\theta_{jm})L_{jm}$, since one unit of developer value is worth one to her. The first-order condition is $1 - \theta_{jm}(1 + \partial \ln \theta_{jm}/\partial \ln L_{jm}) = 0$, which gives (8). For the aggregate,

$$\frac{1}{\bar{\theta}_m} = \frac{\sum_j L_{jm}}{\sum_k \theta_{km} L_{km}} = \sum_j \frac{w_{jm}}{\theta_{jm}} = \sum_j w_{jm} \left(1 + \frac{w_{jm}}{\eta^S} + \frac{1-w_{jm}}{\eta^D} \right) = 1 + \frac{HHI_m}{\eta^S} + \frac{1-HHI_m}{\eta^D}.$$

Corollary 1 follows by setting $HHI_m = 1$, letting $HHI_m \to 0$, and letting $\eta^D \to \infty$. □

Lemma 2 Under the perceived process, the standing value satisfies $V_i^w(R) = \mathbb{E}[\int_0^\tau e^{-\rho t}(R_t h_i^{\text{old}} + v_i^C)\,dt] + \mathbb{E}[e^{-\rho\tau} V_i^w(R_\tau)]$ for every stopping time τ. An owner who sells at τ collects the standing flows until τ and receives $\bar{\theta}_m V_i^D(R_\tau)$ at τ. Her value is therefore $V_i^w(R) + \mathbb{E}[e^{-\rho\tau}(\bar{\theta}_m V_i^D - V_i^w)(R_\tau)] = V_i^w(R) + \bar{\theta}_m \mathbb{E}[e^{-\rho\tau} S_i^\theta(R_\tau)]$. Taking the supremum over τ gives (10). □

Proposition 2 Let $\mathcal{Q}(x) \equiv \frac{1}{2}\sigma^2 x(x-1) + \delta x - \rho$, with roots $\beta_- < 0 < 1 < \beta$. In the continuation region, the option value solves $\frac{1}{2}\sigma^2 R^2 \mathcal{O}'' + \delta R \mathcal{O}' - \rho \mathcal{O} = 0$, so $\mathcal{O}_i^\theta = C_1 R^\beta + C_2 R^{\beta_-}$, and boundedness at zero imposes $C_2 = 0$. Value matching, $C_1 R^{\star\beta} = S_i^\theta(R^\star)$, and smooth pasting, $\beta C_1 R^{\star\beta-1} = S_i^{\theta\prime}(R^\star)$, give the condition $\beta S_i^\theta = R S_i^{\theta\prime}$. In the constrained regime, $V_i^D(R) = q_i R/(\rho-\delta) - (F_i + C_i + \tau^A \bar{h}_i)/(1+\tau^M)$ and $V_i^w(R)/\bar{\theta}_m = [R h_i^{\text{old}}/(\rho-\delta) + v_i^C/\rho]/\bar{\theta}_m$, which gives a_i^θ and K_i^θ. The condition reads $\beta(a_i^\theta R - K_i^\theta) = a_i^\theta R$, which gives (11). In the interior regime without standing housing and with $\tau^A = 0$, $V_i^D = b_i R^{\zeta+1} - F_i/(1+\tau^M)$ and the condition reads $\beta(b_i R^{\zeta+1} - K_i^{\text{int},\theta}) = (\zeta+1) b_i R^{\zeta+1}$, which gives (12). The constrained payoff is affine, so its option value is finite for any $\beta > 1$. A parcel that stays in the interior regime requires $\beta > \zeta + 1$, since $\mathbb{E}[e^{-\rho t} R_t^n]$ is bounded if and only if $n < \beta$. □

Proposition 3 (i) In the constrained regime, $K_i^\theta/a_i^\theta = (\rho-\delta)[(F_i+C_i+\tau^A\bar{h}_i)/(1+\tau^M)+v_i^C/(\bar{\theta}_m\rho)]/[q_i - h_i^{\text{old}}/\bar{\theta}_m]$. The numerator decreases with $\bar{\theta}_m$, strictly if $v_i^C > 0$. The denominator increases with $\bar{\theta}_m$, strictly if $h_i^{\text{old}} > 0$. The threshold therefore decreases with $\bar{\theta}_m$, strictly if $V_i^w > 0$, and does not depend on $\bar{\theta}_m$ if $V_i^w = 0$. In the interior regime without standing housing, $K_i^{\text{int},\theta}$ decreases with $\bar{\theta}_m$, strictly if $v_i^C > 0$. Since $\bar{\theta}_m \le 1$, $R_i^\star \ge R_i^{\text{SP}}$. Parcels whose threshold straddles the saturation rent are solved numerically, and the property holds there by the same argument, since $S_i^\theta = S_i - (1/\bar{\theta}_m - 1)V_i^w$ falls pointwise as $\bar{\theta}_m$ falls. (ii) The planner redevelops the parcel if $a_i = (q_i - h_i^{\text{old}})/(\rho-\delta) > 0$, that is if $h_i^{\text{old}} < q_i$. Privately, $a_i^\theta \le 0$ if $\bar{\theta}_m \le h_i^{\text{old}}/q_i$. Then $S_i^\theta(R) = a_i^\theta R - K_i^\theta < 0$ at every rent, since $K_i^\theta > 0$, and the owner never sells. □

Proposition 4 With zero taxes, total surplus on parcel i under a threshold $r \ge R_0$ is $\text{TS}_i(r) = V_i^w(R_0) + S_i(r)(R_0/r)^\beta$. The developer's rent is a transfer within this sum. The planner's threshold R_i^{SP} maximises $S_i(r)r^{-\beta}$, which is single-peaked on the region where $S_i > 0$. (i) For a parcel with a standing value, $R_i^\star > R_i^{\text{SP}}$ by Proposition 3, so $\partial\text{TS}_i/\partial r < 0$ at $R_i^\star$. Since $R_i^\star$ decreases with $\bar{\theta}_m$, $d\text{TS}_i/d\bar{\theta}_m = (\partial\text{TS}_i/\partial r)(\partial R_i^\star/\partial\bar{\theta}_m) > 0$, a first-order gain. For a parcel blocked by concentration, raising $\bar{\theta}_m$ above h_i^{old}/q_i yields a discrete gain. (ii) If $\bar{\theta}_m = 1$, the private threshold is R_i^{SP}. A subsidy s moves it to $r(s) = R_i^{\text{SP}}(F_i - s)$, while the resource cost remains F_i. Since $r(0)$ maximises TS_i, $d\text{TS}_i/ds|_{s=0} = (\partial\text{TS}_i/\partial r)\, r'(0) = 0$, and the change in total surplus is $O(s^2)$. The outlay moves resources from taxpayers to the parcel and changes welfare only through the shadow cost λ_g. (iii) With a subsidy at rate τ on the land price, the owner receives $(1+\tau)\bar{\theta}_m V_i^D$ on sale. By the argument of Lemma 2, she exercises on $V_i^D - V_i^w/[(1+\tau)\bar{\theta}_m]$, which equals S_i when $(1+\tau)\bar{\theta}_m = 1$. Her threshold is then R_i^{SP}. The subsidy multiplies every offer by $1+\tau$, so parcel supply (7) depends on $(1+\tau)\theta_{jm}$ and $(1+\tau)\Theta_m$. The elasticity $\partial \ln L_{jm}/\partial \ln \theta_{jm}$ is unchanged, and so are the first-order condition, the offer θ_{jm}, and the developer's rent per unit of developer value. (iv) A lump-sum subsidy s_i shifts the private surplus to $S_i^\theta + s_i$. In the constrained regime, the threshold becomes $\frac{\beta}{\beta-1}(K_i^\theta - s_i)/a_i^\theta$, which equals R_i^{SP} only for $s_i = K_i^\theta - K_i a_i^\theta/a_i$, a function of the fixed cost and the standing value of each parcel. At $\bar{\theta}_m < 1$, a parcel with a standing value has $R_i^\star > R_i^{\text{SP}}$, so $\partial\text{TS}_i/\partial r < 0$ at $R_i^\star$, and a small uniform subsidy that lowers its threshold raises total surplus to first order. A parcel without one has $R_i^\star = R_i^{\text{SP}}$, which maximises TS_i, so the same subsidy lowers total surplus to second order, as in (ii). If $a_i^\theta \le 0$ and $s_i \le F_i$, the private surplus is $a_i^\theta R - (K_i^\theta - s_i)$ with $K_i^\theta - s_i \ge C_i + \tau^A\bar{h}_i + v_i^C/(\bar{\theta}_m\rho) > 0$, so it is negative at every rent. Finally, $HHI_m \to 0$ gives $\bar{\theta}_m \to \theta^{\text{atom}} < 1$, and (i) applies. □

Proposition 5 For a fixed stopping time τ, $\mathbb{E}[e^{-\rho\tau}S_i^\theta(R_\tau; F)]$ is affine and strictly decreasing in F, with slope $-\mathbb{E}[e^{-\rho\tau}]/(1+\tau^M) < 0$ whenever τ is finite with positive probability. The supremum of a family of continuous decreasing functions is continuous and decreasing, and strictly so because the optimal stopping time is finite with positive probability for every finite F. As $F \to \infty$, the threshold diverges. Since the payoff is eventually affine, $\mathcal{O}_i^\theta(R_{i0}; F) \le a_i^\theta R_{i0}^\beta (R_i^\star)^{1-\beta} \to 0$ because $\beta > 1$. The map $F \mapsto \mathcal{O}_i^\theta(R_{i0}; F)$ is therefore a continuous, strictly decreasing bijection from $[0,\infty)$ onto $(0, \mathcal{O}_i^\theta(R_{i0}; 0)]$. The observed price of an unsold parcel is its value to the owner, $\hat{p}_i^L = V_i^w + \bar{\theta}_m\mathcal{O}_i^\theta(R_{i0}; F_i)$ by Lemma 2, so a unique $F_i \ge 0$ exists if and only if (14) holds. □

Proposition 6 By Proposition 3, $R_i^\star \ge R_i^{\text{SP}}$ for every parcel. On every demand path, the running maximum is nondecreasing, so the start date $\inf\{t : \phi_m\mathcal{M}_t \ge R_i^\star\}$ is weakly later under the markdown. The set of parcels started by any date is therefore weakly smaller. Parcels that build at the cap deliver $\bar{h}_i - h_i^{\text{old}}$ whenever they start, so delivered floorspace is weakly lower. If $\mathcal{M}_t \to \infty$, every parcel with a finite threshold eventually starts under both regimes, and the gap converges to the floorspace of parcels with $a_i^\theta \le 0 < a_i$. □

Proposition 7 (i) In the constrained regime,

$$r_i = \frac{1}{\phi_m}\,\frac{\beta}{\beta-1}\,\frac{(\rho-\delta)\left[F_i + C_i + \tau^A \bar{h}_i + (1+\tau^M) v_i^C/(\bar{\theta}_m \rho)\right]}{(1-\tilde{t}^F)\lambda_i^R \bar{h}_i/(1+\tau^V) - (1+\tau^M) h_i^{\text{old}}/\bar{\theta}_m},$$

with a positive denominator. The numerator increases with F_i, τ^A, τ^M, and v_i^C. The denominator decreases with τ^V, $\tilde{t}^F$, τ^M, and h_i^{old}. Both move against redevelopment as $\bar{\theta}_m$ falls (Proposition 3). The premium $\beta/(\beta-1)$ enters multiplicatively. For the lag, divide numerator and denominator by λ_i^R. The denominator falls with ℓ_i. The numerator rises, since $C_i/\lambda_i^R = c_0 T_i^{-1/\zeta} \bar{h}_i^{(\zeta+1)/\zeta}/\kappa_i$ and κ_i decreases with ℓ_i when $\delta < \rho/2$: indeed $d\ln\kappa_i/d\ell_i = -(\rho-\delta) + 1/\ell_i - \rho/(e^{\rho\ell_i}-1) \leq \delta - \rho/2$. In the interior regime without standing housing, $(\phi_m r_i)^{\zeta+1} = \frac{\beta}{\beta-\zeta-1}[F_i + (1+\tau^M) v_i^C/(\bar{\theta}_m\rho)]/b_i^0$, where $b_i^0 = (1+\tau^M) b_i$ is proportional to $[(1-\tilde{t}^F)/(1+\tau^V)]^{\zeta+1}(\lambda_i^R)^{\zeta+1}/(\lambda_i^C)^{\zeta}$. The same monotonicity follows, and $d\ln b_i^0/d\ell_i \leq -(\zeta+1)(\rho-\delta) + \zeta\rho/2 < 0$ when $\delta < \rho/2$. (ii) By (3), $h_i^\star(R)$ depends on R, T_i, c_0, ζ, $\rho-\delta$, and $\hat{\kappa}_i = \kappa_i(1-\tilde{t}^F)/(1+\tau^V)$, and increases with $\hat{\kappa}_i$. None of F_i, h_i^{old}, v_i^C, β, τ^M, or $\bar{\theta}_m$ enters it, and τ^A does not by the treatment of the development tax. (iii) Delivery occurs ℓ_i years after the start, with ℓ_i deterministic by Assumption 3. □

Proposition 8 In the constrained regime, $\ln R_i^\star = \ln[\beta/(\beta-1)] + \ln K_i^\theta - \ln a_i^\theta$. By Proposition 2, $(1+\tau^M) K_i^\theta = F_i + C_i + \tau^A \bar{h}_i + (1+\tau^M) v_i^C/(\bar{\theta}_m\rho)$. Since $C_i \propto \bar{h}_i^{(\zeta+1)/\zeta}$, $\bar{h}_i\, dC_i/d\bar{h}_i = (1+1/\zeta) C_i$, so $d\ln K_i^\theta / d\ln \bar{h}_i = \varepsilon_i^K$. Since $q_i \propto \bar{h}_i$, $d\ln a_i^\theta / d\ln \bar{h}_i = q_i/(q_i - h_i^{\text{old}}/\bar{\theta}_m) = \varepsilon_i^a$, which is at least one, and equal to one if and only if $h_i^{\text{old}} = 0$. Finally, $\varepsilon_i^K \leq 1$ if and only if $C_i/\zeta \leq F_i + (1+\tau^M) v_i^C/(\bar{\theta}_m\rho)$. On a parcel without standing value, $\varepsilon_i^a = 1$ and $v_i^C = 0$, so the threshold rises if and only if $C_i/\zeta > F_i$. □

Proposition 9 (i) If $r_i = (1+\omega) r_i^{\text{SP}}$ for every parcel, the set $\{i : r_i \leq r\}$ equals $\{i : r_i^{\text{SP}} \leq r/(1+\omega)\}$ for every regional rent r. The private economy converts at rent r exactly the parcels the planner converts at $r/(1+\omega)$, so the order of conversion and the set of converted parcels for any quantity of new floorspace coincide. (ii) In the constrained regime, Proposition 2 gives $R_i^\star/R_i^{\text{SP}} = (K_i^\theta/K_i)(a_i/a_i^\theta)$, with $K_i^\theta = K_i + (1/\bar{\theta}_m - 1) v_i^C/\rho$, $a_i^\theta = a_i - (1/\bar{\theta}_m - 1) h_i^{\text{old}}/(\rho-\delta)$, and $a_i = (q_i - h_i^{\text{old}})/(\rho-\delta)$. Substituting gives (18). Both the numerator and the inverse of the denominator equal one if $v_i^C = h_i^{\text{old}} = 0$, and both increase with $1/\bar{\theta}_m - 1$. (iii) The fixed cost enters K_i^θ and K_i, the lag enters q_i and C_i, and the option premium multiplies both thresholds. They therefore enter the planner's threshold as well as the private one. Through K_i and q_i they affect the size of the markdown wedge, but they are not wedges themselves. □

C. Calibration appendix

C.1. Demand

I reconstruct $\log \hat{A}_t = \log R_t + \alpha \log Q_t$ annually over 2015–2025, with R_t the regional hedonic rent index and Q_t the housing stock in square metres. Under Assumption 1, $\Delta \log A_t = (\mu - \sigma^2/2) + \sigma\epsilon_t$ with independent standard normal draws ϵ_t. The estimators are $\hat{\sigma}^2 = \widehat{\mathrm{Var}}(\Delta \log \hat{A}_t)$ and $\hat{\mu} = \overline{\Delta \log \hat{A}_t} + \hat{\sigma}^2/2$, with block-bootstrap standard errors (Table C1). I use $\hat{\sigma}$ and set the drift by hand to $\mu = 1.5\%$, against an estimate of $\hat{\mu} = 2.2\%$. The series contains only 10 annual changes, so $\hat{\sigma}$ is imprecise. Smoothed rent indices also bias $\hat{\sigma}$ down, which raises β and lowers every threshold. Appendix Table E4 reports how the results move with α, which changes $\hat{\sigma}$.

Table C1: Demand shifter reconstruction

Year	Rent index	Stock (million m2)	Quality-adjusted stock (million m2)	log A	dlog A
2015	17.052	211.807	266.921	31.795	–
2016	17.325	213.261	268.478	31.820	0.025
2017	17.326	214.485	269.792	31.827	0.007
2018	17.328	215.759	271.146	31.835	0.008
2019	17.551	217.118	272.613	31.855	0.021
2020	17.933	218.543	274.143	31.885	0.030
2021	18.184	220.225	275.904	31.909	0.023
2022	18.264	222.222	277.962	31.924	0.015
2023	18.726	223.797	279.572	31.958	0.034
2024	19.052	225.398	281.197	31.984	0.026
2025	19.350	227.714	283.570	32.012	0.028
mu, estimate (%)	–	–	–	–	2.172
s.e. (%)	–	–	–	–	0.315
mu, used (%)	–	–	–	–	1.500
sigma (%)	–	–	–	–	0.897
s.e. (%)	–	–	–	–	0.136
sigma, price index (%)	–	–	–	–	3.394

Data: *Observatoire des Loyers*, *Fichiers Fonciers*, DVF. Annual $\hat{A}_t$ at $\alpha = 1.49$. Bottom rows: estimated drift $\hat{\mu}$ with its bootstrap standard error, the drift used, $\hat{\sigma}$ with its bootstrap standard error, and the volatility of a transaction-based price index for comparison.

C.2. Discounting

The discount rate ρ is the sum of a risk-free rate and a development risk premium. The risk-free rate is the average 10-year OAT yield over 2015–2025, 1.27%. The premium is the spread between the post-tax weighted average cost of capital that a listed developer discloses for its residential segment and the OAT yield of the same year. This rate is computed under the CAPM by independent experts for IAS 36 impairment tests. I use the rate disclosed by Nexity, the main listed French developer: 7.9% in 2023 and 8.4% in 2024. The average spread over these two years is 5.17 percentage points. I set $\rho = 6.4\%$, close to the sum of the two. The net residential yield y is annual rent net of property tax, non-recoverable charges, and vacancy, divided by price. I set $y = 4.4\%$. For comparison, the average ratio of municipal rent indicators to DVF prices per square metre is 4.4%. Capitalising municipal rents at y reproduces DVF prices per square metre with a mean absolute deviation of 18.9%. Since the price of a rental flow is $R/(\rho - \delta)$, the perceived drift is $\delta = \rho - y = 2.0\%$. Appendix Table E4 reports how β and the main results move with $\rho \pm 1$ point and $y \pm 50$ basis points. Because β is close to ρ/δ when σ is small, thresholds depend more on the discount rate and the yield than on the volatility of rents.

C.3. Construction

PRLN reports, for each surveyed project j, its construction cost net of demolition and land. Taking logs of the variable-cost function of Assumption 3,

$$\log C_j^{\text{var}} = \log c_0 - \tfrac{1}{\zeta} \log T_j + \left(1 + \tfrac{1}{\zeta}\right) \log h_j + X_j' b + e_j, \tag{C1}$$

with project-type and year controls X_j. The coefficients on log parcel area and log floorspace sum to one, a testable restriction of the Cobb-Douglas technology. I impose ζ from Combes et al. (2021) and estimate $c_0 = 1,186$ from the intercept, at the mean of the controls, in a separate estimation on the PRLN micro-data, accessed through the CASD. The value is set by hand in the pipeline. Costs are contemporaneous with the works and measured net of deductible VAT, so the lag factor λ_i^C is not applied a second time.

C.4. Construction lag

For parcels with a DVF sale followed by a permit, the lag ℓ_u is the median time from commitment to completion, by use of origin. In France the deed is usually signed after the permit, once the conditions of the *promesse de vente* are met (Section 2.1). I therefore date commitment at the earlier of the deed and the permit authorisation. Completion is the declared end of works in *Sitadel* where reported, and otherwise the first *BD Topo* vintage that shows the new building, minus half the interval between vintages.

Table C2: Construction lag by use of origin: two datings compared

use	Transaction to completion, mean	Transaction to completion, median	Permit to completion, median	Administrative gap, median	N
housing	38.3	32.7	40.0	-6.5	1,505.0
mixed	40.3	33.4	41.0	-6.7	920.0
nonhousing	49.8	40.0	39.0	-6.0	37.0
vacant	39.8	27.6	50.0	-10.8	7.0

Data: DVF, *Sitadel*, *BD Topo*. Mean and median months, by use of origin, from (i) DVF transaction to completion and (ii) permit to completion, the dating of Fact 3, with the implied administrative gap. A negative gap means that the deed follows the permit. The lag ℓ_u starts at the earlier of the two dates.

C.5. Taxes

This appendix details the four taxes of Section 3.2 and the levies left out of the model. Rates are those in force in 2025.

Value-added tax New dwellings sold by a developer are subject to VAT at the standard rate of 20% on the full sale price (article 278 of the *Code général des impôts*, CGI). The developer deducts the VAT paid on construction works, so VAT is a proportional wedge on the value of the completed building. Reduced rates apply to owner-occupied housing sold in urban renewal zones (5.5%) and to intermediate rental housing (10%). These programmes are a small share of private permits, and I apply the standard rate to every project. When the land is bought from an owner who is not subject to VAT, no deductible VAT is attached to it, so VAT also falls on the land component of the price. The wedge $1/(1+\tau^V)$ on the value of the completed building captures this.

Transfer tax The DMTO is paid by the buyer on the price of the land and of any structure on it. It combines a departmental rate, a municipal rate of 1.2%, and a state levy of 2.37% of the departmental amount. The 2025 finance law allows *départements* to raise their rate from 4.5% to 5% between 2025 and 2028. I apply 5%, which gives $\tau^M = 6.32\%$. A developer who buys land subject to VAT and commits to build within four years pays a fixed duty instead (article 1594-0 G of the CGI). Most acquisitions in the sample are purchases of existing structures from households, which do not qualify. The full rate therefore applies to most of them, and it overstates τ^M for the others.

Development tax The *taxe d'aménagement* is due on the floorspace authorised by the permit. Its base is a statutory value per square metre, 1,036 € in Île-de-France in 2025. Three rates apply: a municipal rate between 1% and 5%, which can reach 20% in designated sectors, a departmental rate of at most 2.5%, and a regional rate of at most 1% in Île-de-France. I apply the three standard maxima, 5%, 2.5%, and 1%, which gives $\tau^A = 88$ € per square metre of new floorspace. The tax is paid after the permit, in one or two instalments. I treat it as paid at the start date.

Property tax The *taxe foncière sur les propriétés bâties* is paid annually by the owner. Its base is half the cadastral rental value of the building, and the municipality sets the rate. Since 2021, the municipal rate includes the former departmental share. Cadastral rental values date from 1970 and are revised only by statutory coefficients, so the base is a small fraction ψ^{new} of the market rent. Rates vary across municipalities. I apply the municipal rate t_m^F of the DGFiP file of local tax rates (REI) where it is available, which covers 100% of parcels, and a rate of 30% elsewhere. The base share is $\psi^{\text{new}} = 0.25$. New dwellings are exempt for two years after completion (article 1383 of the CGI). Municipalities may limit the exemption on their share of the base. I apply the full exemption, $e = 2$ years. The standing structure pays the property tax too. The model does not deduct it from the standing value (1), which uses the market rent. It enters the fiscal accounts of Section 6.1, as the receipts lost once the structure is emptied.

Levies left out The household waste levy (TEOM) and the special equipment tax are assessed with the property tax. They fall on old and new buildings alike, so they change the level of both values but not, to first order, the comparison between them. The preventive archaeology levy is small. Connection charges for sewerage (PFAC) depend on the network, and they are part of the fixed cost. The capital gains tax on the owner's sale is the main omission. Gains on a principal residence are exempt. Other gains are taxed, with an abatement that grows with the holding period and exempts the gain after 22 years for income tax and after 30 years for social contributions. The tax lowers the net price a recent owner receives, so it raises her threshold. It acts like the markdown on parcels held for a short time, and it vanishes for parcels held long.

D. Estimating developer market power

This appendix details the calibration of the land share summarised in Section 4.2. The land market is defined at the municipality level. I compute each developer group's share of land payments from the universe of *Sitadel* redevelopment permits, after consolidating project-specific vehicles into their parent group. Each permit is weighted by its floorspace times the local price of new housing, a proxy for the value of the project to the developer. Shares are computed over a rolling three-year window, which smooths lumpy projects.

D.1. The slope: municipal concentration

Equation (20) uses *DVF* sales of housing parcels followed by a permit on the same parcel, filed at most one year before and 3 years after the sale, with the municipal Herfindahl index of the year of the sale. Mixed and non-housing parcels are excluded. For them, the model value g_i misses the value of the existing use, and the price often exceeds g_i. The index is instrumented with its average over the other municipalities of the same EPT. Standard errors are clustered by EPT, with 12 clusters. With so few clusters, the confidence interval uses the Student critical value with one degree of freedom less than the number of clusters, 2.20, rather than 1.96. I also report a wild cluster bootstrap p-value (Cameron et al., 2008). The first stage has a Wald statistic of 52. The mapping $s = b/(1 + b\,\overline{HHI})$ uses the mean index of the estimation sample, $\overline{HHI} = 0.215$, and is exact to first order around it. The slope is $b = -0.52$ (s.e. 0.32), so $s = -0.59$ (s.e. 0.40) with a 95% confidence interval of $[-1.65, 0.17]$ (Table D1). The calibration sets s at the lower bound of this interval. It is the largest markdown the data allow, so the effects of concentration in the counterfactuals are upper bounds. Incomplete consolidation scales measured indices down. If measured indices were proportional to true ones, this would push b away from zero and leave the implied land shares nearly unchanged, since s enters them through $s\,HHI_m$.

D.2. The level: developer margins

The fixed costs absorb the level of the land share in sale prices. I pin it down with the operating margins of listed developers. Given s, the land share in municipality m is $\bar{\theta}_m = \theta^{\text{atom}}/(1 - s\,HHI_m)$, so a single scalar remains. I set θ^{atom} so that the median model-implied operating margin equals the median margin of listed French developers, 8.0%. The margin is computed on every parcel of the estimation sample with a finite threshold, at that threshold $R_i^\star$. The margin of parcel i is the developer's rent, $(1 + \tau^M)(1 - \bar{\theta}_{m(i)})V_i^D(R_i^\star)$, relative to the value of the completed building. Both elasticities then follow from (21). With $s = -1.65$, the margins give $\theta^{\text{atom}} = 100\%$, $\eta^D = 969.47$ and $\eta^S = 0.61$. Through (9), the observed indices imply a land share of 62% on average, between 41% and 78% across municipalities.

D.3. Joint solution for the level and the fixed costs

The fixed costs F_i are recovered by inverting observed land prices, given the land share. The model margin depends on these fixed costs. I solve for θ^{atom} and the fixed costs with a damped fixed point. Given θ^{atom} and s, (21) gives η^D and η^S, and (9) gives $\bar{\theta}_m$ in every municipality. I then recover F_i for every parcel, compute its threshold $R_i^\star$ and its developer value relative to the value of the completed building. Holding these ratios fixed, a new θ^{atom} sets the median margin equal to the target exactly. The update averages the old and new values with equal weights. The procedure runs first on a random sample of 60,000 parcels, then on all parcels, until θ^{atom} changes by less than 10^{-4}.

Table D1: The land market: slope and calibrated parameters

Panel A. Slope of the land share in municipal concentration, Eq. (20)

Statistic	Value
Slope of log land share on HHI, b	-0.520
s.e.	0.317
First stage (Wald)	51.5
Mean HHI	0.215
Implied s	-0.585
s.e.	0.402
Implied s, 95% lower bound (calibration)	-1.650
Implied s, 95% upper bound	0.172
Land sales	5,615
Clusters (EPT)	12
Wild cluster bootstrap p-value of b	–

Panel B. Calibrated land-market parameters, s at its 95% lower bound

Slope s	-1.65
Atomistic land share θ^{atom} (%)	100
Elasticity η^D	969.47
Elasticity η^S	0.61
Payment-weighted regional index $\overline{HHI}$	0.321
Land share, mean (%)	62
Land share, 10th to 90th percentile (%)	41–78

Notes: Panel A estimates (20) on housing land sales followed by a permit on the same parcel, by two-stage least squares with département and year effects. The municipal Herfindahl index is instrumented with its average over the other municipalities of the same EPT. Standard errors clustered by EPT. Panel B sets s at the lower bound of its 95% confidence interval and combines it with the median operating margin of listed developers through (21).

E. Additional results

E.1. Fifty-year horizon

Table E1: Counterfactuals at 20 and 50 years

Scenario	dQ (%)	dQ (50y, %)	dR (%)	dR (50y, %)	Net cost/unit (k€)	Net cost/unit, (50y, k€)
Density: caps raised to the median	0.51	2.15	-0.76	-3.13	-52.03	-33.72
Density: caps lifted	0.27	1.75	-0.40	-2.57	-67.21	-33.39
Fonds Friches scale (50 euros/m2)	0.22	0.12	-0.33	-0.18	-23.77	0.24
Largest self-financing scale (1050 euros/m2)	3.80	2.34	-5.41	-3.39	-0.26	32.92
Competitive land market	1.09	0.68	-1.60	-1.01	-78.53	-98.69
Deconcentration (HHI to zero)	1.08	0.68	-1.60	-1.01	-78.59	-98.77
Subsidy to sellers	1.09	0.68	-1.60	-1.01	99.86	258.98
Mandate and offsetting subsidy	-0.69	-1.05	1.04	1.60	–	–

Data: model estimation. Change in the quality-adjusted stock and in the rent, and net cost per additional unit of 60 square metres, with units and receipts both counted until year 20 and until year 50.

E.2. Fixed-cost subsidy: grid of scales

Table E2: Fixed-cost subsidy: grid of scales

Scale (euros/m2)	dQ (%)	Outlay (m€)	Induced receipts (m€)	Net balance (m€)
900	3.28	9,377.02	9,855.76	478.74
950	3.44	10,040.73	10,340.87	300.14
1,000	3.61	10,716.74	10,878.48	161.73
1,050	3.80	11,416.68	11,461.49	44.80
1,100	3.96	12,136.61	12,008.85	-127.76
1,150	4.13	12,859.60	12,516.10	-343.50
1,200	4.31	13,570.90	13,029.00	-541.90
1,250	4.49	14,264.01	13,543.98	-720.03
1,300	4.64	15,042.33	14,027.09	-1,015.24
1,350	4.81	15,888.27	14,540.54	-1,347.73
1,400	4.98	16,661.48	15,042.83	-1,618.65
1,450	5.10	17,458.79	15,506.71	-1,952.08
1,500	5.23	18,254.53	15,989.75	-2,264.78

Data: model estimation. Subsidy of s euros per square metre of parcel area, capped at the fixed cost, on a grid from 900 to 1,500 euros in steps of 50. Columns: change in the quality-adjusted stock after 20 years, outlay, tax receipts induced relative to the private equilibrium, and net balance of the consolidated budget, all by year 20. The scale of Section 6.3 is the largest with a non-negative balance.

E.3. A tax on under-used land

An annual tax at rate τ^U on the forgone rent of the shortfall between a fraction χ of the density cap and the standing floorspace, paid until the parcel is redeveloped, lowers the standing value by the present value of that flow:

$$V_i^{w,\tau}(R) = V_i^w(R) - \tau^U \frac{(\chi \bar{h}_i - h_i^{\text{old}})^+ R}{\rho - \delta}. \tag{E1}$$

It raises the surplus by the same amount, lowers the threshold of every under-used parcel, and does not distort the density choice. Unlike a fixed-cost subsidy, it acts on the standing value, which is where the markdown bites. It is a targeted version of the tax on standing structures of Remark 4, and it raises

revenue. French law already allows municipalities in tight housing markets to raise the property-tax base of undeveloped constructible land (article 1396 of the CGI). The tax considered here extends the idea to built parcels whose floorspace falls short of the cap.

Table E3: A tax on under-used land

Tax rate (%)	dQ, 20 y (%)	dR, 20 y (%)	Time saved (y)	Idle parcels redeveloped by year 20	Revenue, years 1-10 (m€/yr)	Revenue, years 11-20 (m€/yr)
1.00	0.52	-0.77	1.83	436.77	315.53	343.87
2.00	1.05	-1.55	3.75	872.63	619.84	673.01
5.00	2.31	-3.35	14.50	2,114.27	1,458.46	1,587.76

Data: model estimation. Annual tax at rate $\tau^U \in \{1\%, 2\%, 5\%\}$ on $R_{it}(\chi \bar{h}_i - h_i^{\text{old}})^+$ with $\chi = 0.5$. Columns: change in the quality-adjusted stock and in the rent after 20 years, years saved, additional idle parcels started by year 20 (expected number), and annual revenue in the first and second decades.

E.4. Robustness

Table E4 reports the main results under alternative calibrations. Every variant re-inverts the fixed costs of the idle stock at its own parameters (Proposition 5), then solves the private equilibrium and the competitive land market on the same demand draws. The first two variants change the slope s: to its point estimate, and to zero, so that concentration has no effect on the land share. In both, θ^{atom} is re-solved on the developer margins, holding the fixed costs at the baseline fixed point. The next three lower the inverse demand elasticity α, which changes the volatility σ of the demand shifter. The drift μ keeps its value. The last four move the discount rate ρ and the net yield y. Outside the first two rows, land shares keep their baseline values.

Table E4: Robustness

Variant	β	Mean land share	Idle parcels with a fixed cost	Blocked by concentration	Median threshold / rent	Policy (c): dQ, 20 y (%)	Policy (c): dR, 20 y (%)	Policy (c): time saved (y)
Baseline	3.19	0.62	53,642	112,299	1.96	1.09	-1.60	3.75
Slope s at its point estimate	3.19	0.75	63,136	67,401	2.02	0.77	-1.13	2.75
No effect of concentration ($s = 0$)	3.19	0.79	63,833	57,967	2.04	0.65	-0.97	2.67
$\alpha = 0.35$	3.19	0.62	53,634	112,299	1.96	1.36	-0.47	3.67
$\alpha = 0.50$	3.19	0.62	53,635	112,299	1.96	1.32	-0.65	3.67
$\alpha = 0.80$	3.19	0.62	53,636	112,299	1.96	1.27	-1.00	3.67
ρ + 1 point	2.46	0.62	62,880	112,299	3.08	0.66	-0.98	7.50
ρ - 1 point	5.31	0.62	43,925	112,299	1.18	0.74	-1.09	1.92
y + 50 bp	4.23	0.62	36,804	113,849	1.32	0.81	-1.20	2.08
y - 50 bp	2.55	0.62	69,393	110,712	2.92	0.74	-1.10	7.17

Data: model estimation. Columns: root β of the value of waiting, mean land share across municipalities, idle parcels whose fixed cost is recovered, idle parcels blocked by concentration, median threshold relative to the current rent, and the change in the quality-adjusted stock and in the rent after 20 years and the years saved under policy (c).

E.5. Regimes, supply paths and maps

Table E5: Regime of parcels at their redevelopment threshold

Use of origin	Blocked by concentration	Blocked by the cap	Constrained	Interior	No fixed cost recovered
Housing	22.3	11.5	10.1	1.0	55.1
Mixed	25.1	9.9	5.8	1.8	57.5
Non-housing	0.0	0.0	35.2	10.5	54.3
Vacant	0.2	0.3	85.2	3.2	11.1

Data: model estimation. Share of parcels of each use of origin, in %: blocked by concentration, blocked by the cap, never redeveloped for another reason, in the constrained regime (density at the cap at the threshold), in the interior regime (density below the cap at the threshold), and without a recovered fixed cost (outside the bounds of (14)). Rows sum to 100.

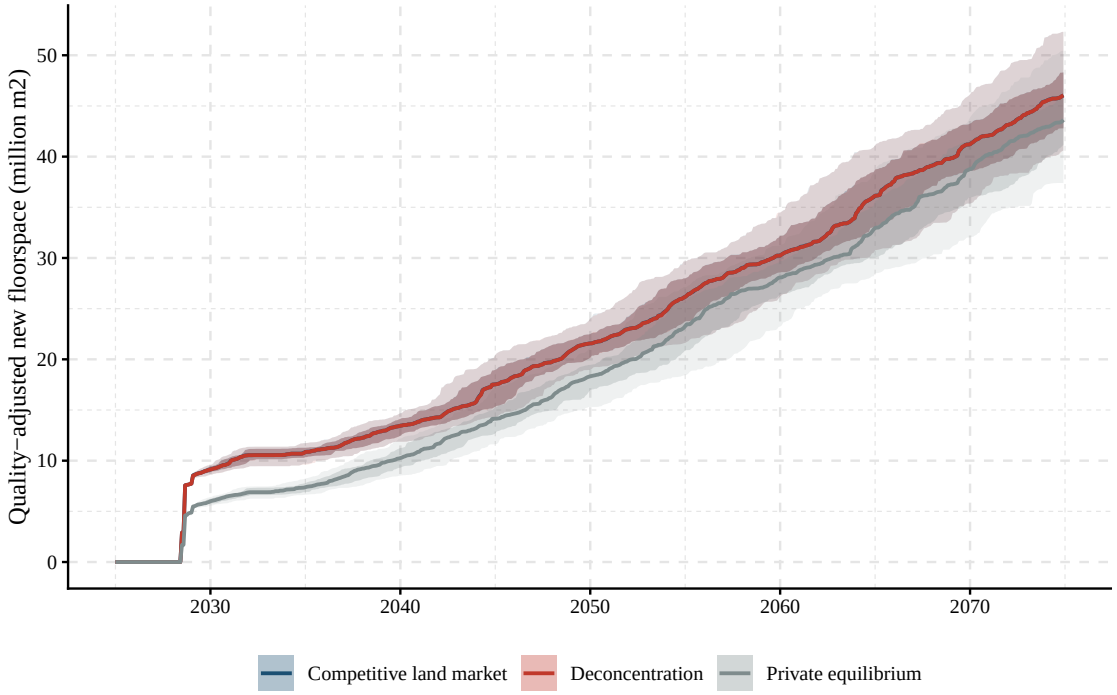


Figure E1: Simulated supply paths: private equilibrium, competitive land market, deconcentration

Data: model estimation. Quality-adjusted new floorspace at the median demand path, with interquartile and interdecile ribbons across simulated draws of A_t, under the private equilibrium, the competitive land market, and deconcentration (Section 6.4).

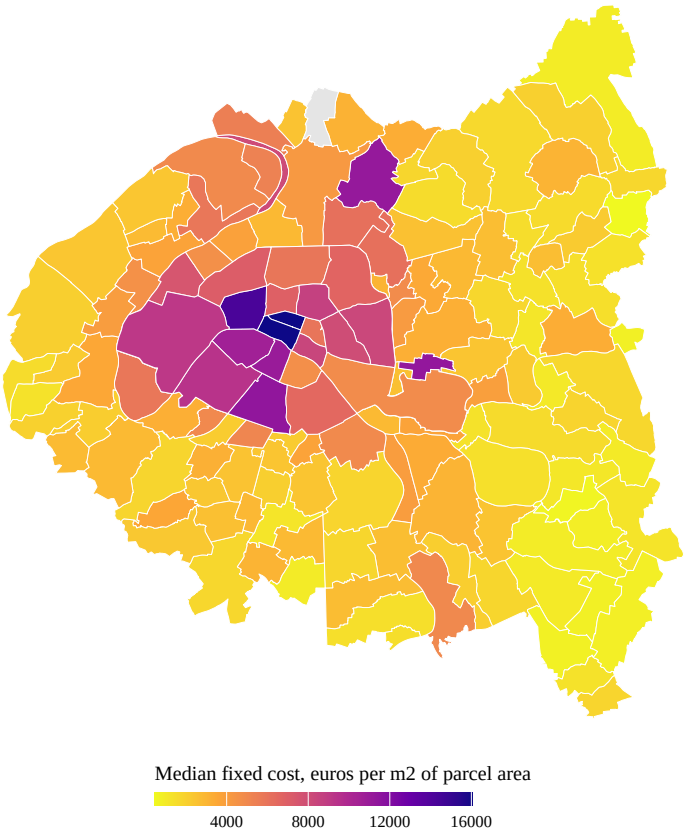


Figure E2: Fixed cost per square metre, by municipality

Data: model estimation. Median recovered fixed cost per square metre of parcel area, by municipality, parcels with a positive fixed cost.

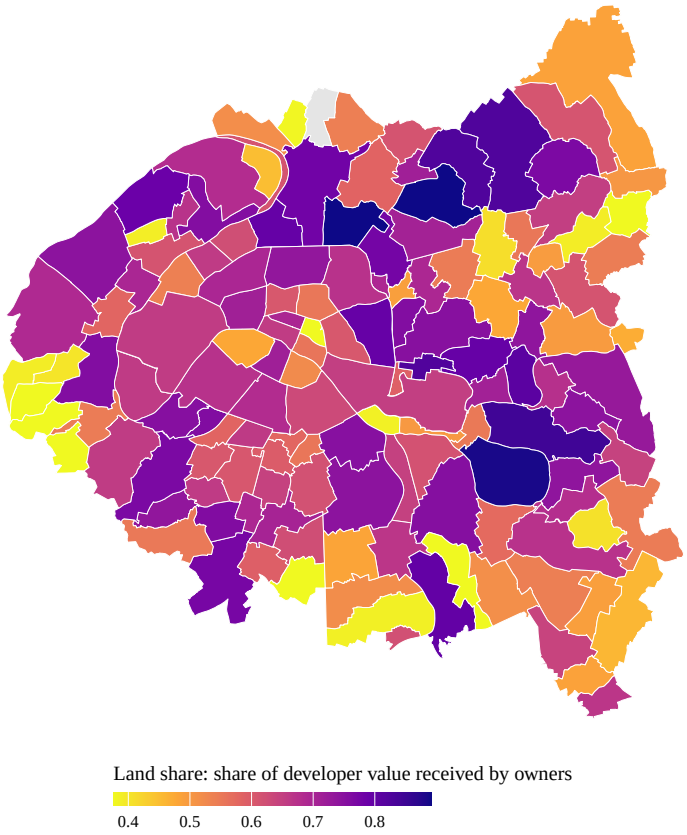


Figure E3: Land share $\bar{\theta}_m$, by municipality

Data: model estimation, *Sitadel*. Share of developer value received by owners, from equation (9) at the estimated elasticities and the municipal Herfindahl index.

F. Composition of the fixed cost

The fixed cost F_i is a residual (Section 3.7). This appendix asks how much of it observable characteristics explain. The sample contains the parcels of the estimation sample with a positive recovered cost. I regress the log fixed cost per square metre of parcel area on eight blocks of observables. *Demolition*: demolished floorspace per square metre of parcel area, the demolition cost predicted from the PRLN survey per square metre, and the height and age of the structure. *Ground*: the BRGM ground class. Networks: connection to the gas or underground electricity network within 100 metres. *Neighbours*: the number of contiguous parcels and their mean height. *Ownership*: the number of owners and a co-ownership indicator. *Area*: log parcel area. *Regulation*: the municipal refusal rate of permits and architectural protection. *Municipal*: population density, income, and distance to Paris. The predicted demolition cost accounts for a median 3.1% of the recovered fixed cost, and 0.2% in aggregate.

I apportion the R^2 of the regression, 0.14, across the eight blocks with a grouped Shapley decomposition: each block receives its average marginal contribution to the R^2 over every order in which blocks could be added (Table F1). The second column demeans every variable by municipality and drops the municipal block. Table F2 reports the share of the fixed cost that each block accounts for, relative to a reference parcel with nothing to demolish, the most frequent ground class, one owner, and every other characteristic at its median.

Two features of the residual matter for policy. The recovered fixed cost depends on the land share: given a price, a lower land share means the owner keeps a smaller share of the option, so the option must be worth more and the fixed cost is lower. And if the share of variance left unexplained is large, part of the residual may reflect a failure inside the fixed cost itself, such as co-ownership hold-outs or the premium of multi-parcel assembly, rather than a technological cost. This decomposition is preliminary.

Table F1: Fixed cost: grouped Shapley shares of the explained variance

Block	All blocks (%)	Within municipality (%)
Demolition	10.50	9.66
Ground	12.76	14.41
Networks	0.52	0.66
Neighbours	16.20	18.25
Ownership	5.08	9.63
Area	16.60	39.05
Regulation	9.29	8.33
Municipal	29.05	–
R^2	0.14	0.07

Data: model estimation, *Fichiers Fonciers*, *BD Topo*, BRGM, PRLN. Dependent variable: $\log(F_i/T_i)$. Each cell is the share of the R^2 attributed to a block (%). Last row: R^2. Within municipality: all variables demeaned by municipality.

Table F2: Fixed cost: decomposition of the level relative to a reference parcel

Block	Median share of F_i (%)	Aggregate share (%)	Median euros per m^2
Demolition	-9.55	-11.94	-179.56
Ground	0.00	1.20	0.00
Networks	0.00	1.88	0.00
Neighbours	0.73	24.52	5.06
Ownership	0.07	-13.94	0.64
Area	-0.00	47.89	0.00
Regulation	0.06	12.51	0.62
Municipal	-3.91	3.05	-18.19

Data: model estimation. For each block, the share of F_i the parcel would save with the characteristics of the reference parcel, $1 - \exp(-\hat{b}'(X_i - X^{\text{ref}}))$: median across parcels, aggregate share of total fixed costs, and median in euros per square metre of parcel area.

G. Tables

Table G1: Housing supply intensity by project type and previous use

Previous destination	Floorspace	Units	FAR	UAR	Parcels
New unit					
New unit	152.0	1	55.11	0.45	1
Within building					
Annex	149.0	1	61.59	0.48	1
Mixed use	103.0	2	21.21	0.47	1
Multi-family unit	117.0	2	38.90	0.88	1
Secondary activity	260.0	5	30.09	1.19	1
Single-family unit	141.0	1	42.93	0.37	1
Tertiary activity	263.0	7	57.77	1.50	1
Vacant	154.0	1	56.44	0.64	1
Demolition					
Annex	153.0	1	54.07	0.42	1
Mixed use	1141.5	19	117.52	1.96	1
Multi-family unit	253.5	3	78.00	1.19	1
Secondary activity	2330.0	35	124.13	1.92	1
Single-family unit	191.0	1	52.67	0.41	1
Tertiary activity	1439.0	26	176.10	3.05	1
Vacant	191.5	1	57.77	0.48	1

Data: *Fichiers Fonciers*, *BD Topo*, *Sitadel*. Floorspace-to-area (FAR×100) and units-to-area (UAR×100) ratios, median unit and floorspace counts, median parcel count, by project type (new unit, within building, demolition) and by previous destination (annex, mixed use, multi-family unit, secondary activity, single-family unit, tertiary activity, vacant).

Table G2: Construction time across uses of origin: Kruskal-Wallis and pairwise Wilcoxon tests

	Annex	Mixed	Multi	New unit	No unit	Secondary	Single
Mixed	0.383						
Multi	0.620	0.058^{*}					
New unit	0.258	$<0.001^{***}$	0.602				
No unit	0.804	0.089^{*}	0.316	0.003^{***}			
Secondary	0.038^{**}	0.051^{*}	0.010^{**}	$<0.001^{***}$	0.020^{**}		
Single	0.634	$<0.001^{***}$	0.804	0.089^{*}	0.049^{**}	0.003^{***}	
Tertiary	$<0.001^{***}$	$<0.001^{***}$	$<0.001^{***}$	$<0.001^{***}$	$<0.001^{***}$	0.708	$<0.001^{***}$
Kruskal–Wallis: $\chi^2 = 118.7$, df $= 7$, $p < 2.2 \times 10^{-16}$							

Data: *Sitadel*. Kruskal-Wallis test of the joint null that construction time is equally distributed across uses of origin. Benjamini-Hochberg adjusted p-values from pairwise Wilcoxon rank-sum tests between each pair of uses.

Table G3: Developer concentration

Panel	Statistic	Value
A. Region	Top 1 share (%)	4.7
A. Region	Top 4 share (%)	11.4
A. Region	Top 7 share (%)	16.1
A. Region	Top 20 share (%)	27.2
A. Region	HHI	0.007
A. Region	1/HHI	153.4
A. Region	N^{tot}, all developers	3977
A. Region	N^{tot}, at least 2 projects	1928
A. Region	N^{tot}, at least 5 projects	469
B. Municipalities (P10 / median / P90)	Municipal HHI	0.17 / 0.33 / 0.89
B. Municipalities (P10 / median / P90)	Effective number of developer groups	1.13 / 3.01 / 5.79
B. Municipalities (P10 / median / P90)	Share of the largest group (%)	27.70 / 49.89 / 94.08
B. Municipalities (P10 / median / P90)	Municipalities	137

Data: *Sitadel*, developer SIREN identifiers, 2013–2024. Panel A, region: cumulative share of redeveloped floorspace by the top 1, 4, 7, and 20 developers, Herfindahl index, effective number of developers, and number of active developers, in total and restricted to developers with at least two and at least five projects. Panel B, municipalities: distribution of the municipal Herfindahl index, of the effective number of developer groups, and of the share of the largest group.

Table G4: Validation

Panel	Statistic	Value
A	Hit rate among parcels predicted ripe (%)	0.93
A	Base rate (%)	1.80
A	Area under ROC curve	0.46
B	Model supply elasticity	0.25
B	Supply elasticity, Paris area, Chapelle et al. (2023)	0.35
B	Model new floorspace, estimation sample (million m2/y)	0.55
B	Sitadel floorspace, same parcels (million m2/y)	0.16
B	Sitadel floorspace, all parcels (million m2/y)	1.48
B	Model supply elasticity, scaled to observed flow	0.68
B	Perceived drift delta = rho - y (%)	2.00
B	Realised drift mu - alpha g_Q (%)	0.60
C	Model operating margin (%), targeted	11.02
C	Listed developers (%)	8.00
D	Reduced-form slope of log land share on HHI	-0.52
D	s.e.	0.32
D	Model-implied slope	-1.22
E, HHI quintile 1	Priced above the zero-cost bound (%)	63.51
E, HHI quintile 2	Priced above the zero-cost bound (%)	79.71
E, HHI quintile 3	Priced above the zero-cost bound (%)	74.62
E, HHI quintile 4	Priced above the zero-cost bound (%)	77.39
E, HHI quintile 5	Priced above the zero-cost bound (%)	73.07

Data: model estimation, DVF, *Sitadel*, *BD Topo*. Panel A: hit rate, base rate, and area under the receiver operating characteristic curve of the timing prediction. Panel B: model and external supply elasticities, calibrated δ against the implied drift. Panel C: model-implied and observed developer margins (targeted). Panel D: reduced-form and model-implied slope of the land share in the municipal Herfindahl index. Panel E: share of parcels priced above the zero-cost bound by quintile of the municipal Herfindahl index.

H. Figures

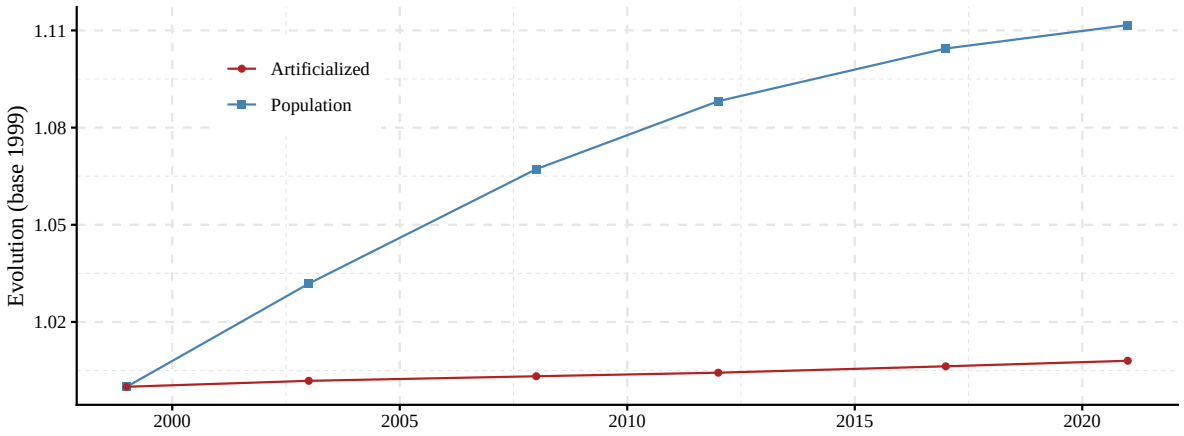


Figure H1: Population and built-up land in the Paris area, 1999–2021

Data: INSEE population censuses. Built-up land from *Fichiers Fonciers*. Indices, 1999 = 1.

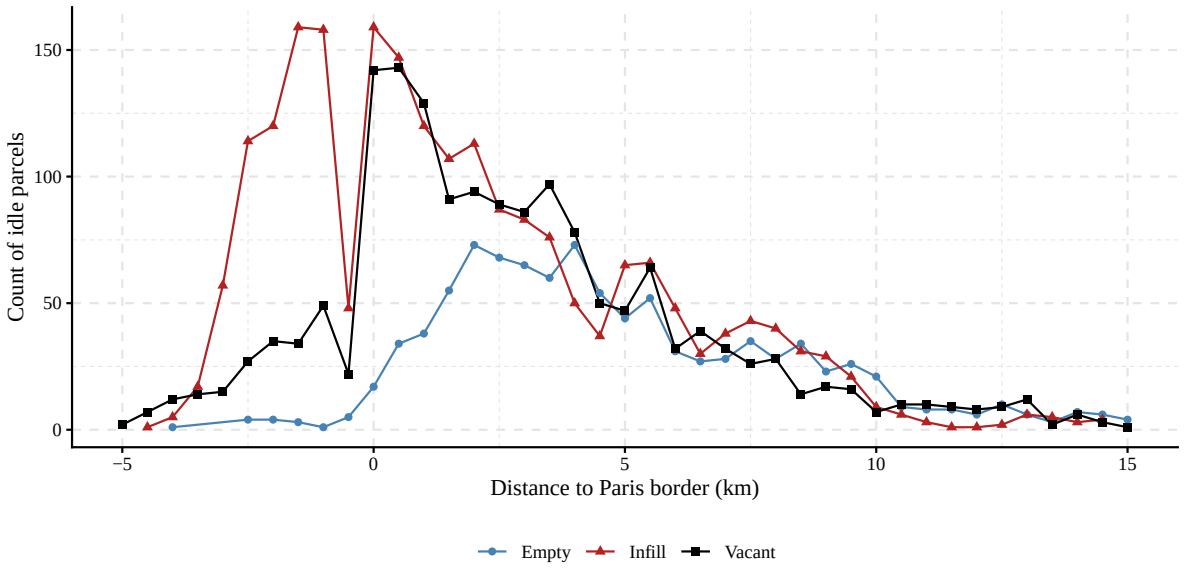


Figure H2: Empty, vacant and infill parcels by distance to Paris

Data: *Fichiers Fonciers*, *Sitadel*. Empty parcels host no physical building. Vacant parcels host a building, no taxed unit, and a documented past economic activity. Infill parcels host a building significantly lower than its neighbours within 100 metres. Data as of 2024.

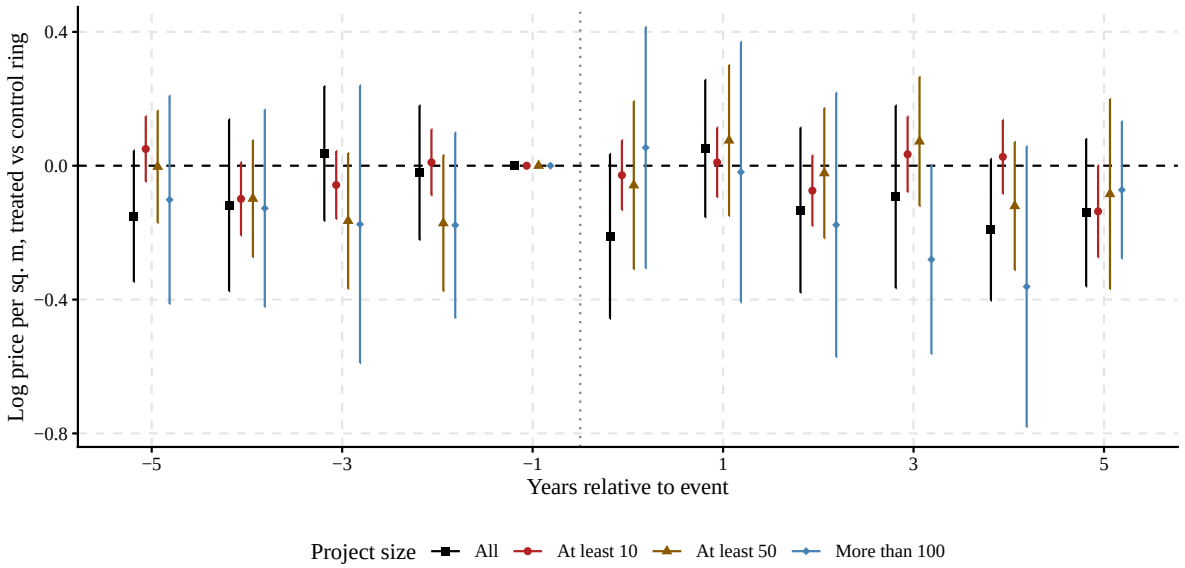


Figure H3: Event study: change in price after delivery

Data: DVF, *Sitadel*, *BD Topo*. Treated ring 20–100 m, buffer 100–150 m (excluded), control ring 150–500 m from the project centroid. Event-time coefficients ξ_k of (A2), relative to the year before delivery ($k = -1$), with confidence intervals. Each colour and shape is a different sample of projects by size.

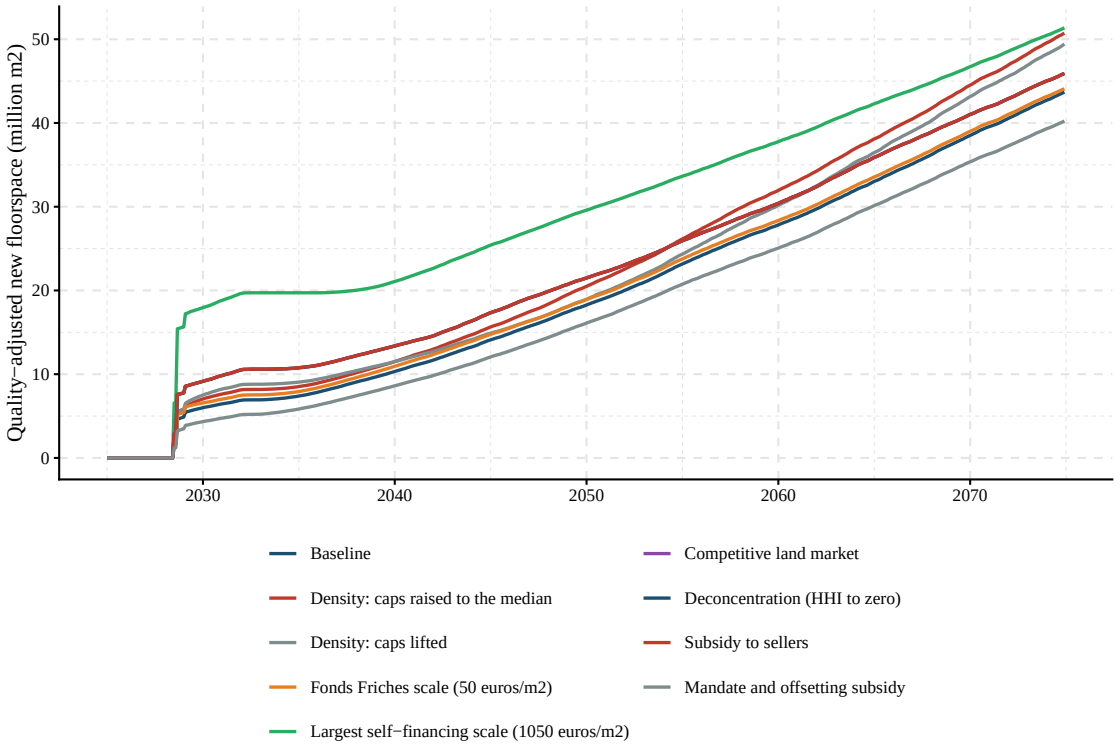


Figure H4: New floorspace under each policy

Data: model estimation. Mean quality-adjusted new floorspace across simulated demand paths, by policy. The same demand paths are used in every scenario.